



Advanced Combustion

— Chapter 1. Introduction

Lei Li (李磊) , Fei Qi (齐飞)

Pre-course



■ About Li Lei (combustion instability)

- 2000 - 2004, Beijing University of Aeronautics and Astronautics, bachelor
- 2004 - 2011, Beijing University of Aeronautics and Astronautics, PhD
- 2011 - 2014, Beijing University of Aeronautics and Astronautics, post-doctor
- 2014 - 2015, Nanyang Technological University, Singapore, post-doctor
- 2016 - now, Shanghai Jiao Tong University

■ About Qi Fei (combustion diagnostics and kinetic modeling)

- 1987 - 1991, Anhui Normal University, bachelor
- 1991 - 1997, University of Science and Technology of China, Ph.D student
- 1997 - 1998, University of Science and Technology of China, Asst./Asso. Prof.
- 1998 - 2001, Lawrence Berkeley National Laboratory, USA, post-doctor
- 2001 - 2003, Sandia National Laboratories, USA, post-doctor
- 2003 - 2014, University of Science and Technology of China, professor
- 2015.01 – now, Shanghai Jiao Tong University, professor

■ About you (name, institute, supervisor, research direction)

Targets of The Course



- Understand the fundamental knowledge of combustion
- Be familiar with basic combustion theory
- Be possible to remember some information of experimental methods when you face these apparatuses
- Know the state-of-the-art research method in this field
- Be aware of the current or potential use of this tool in your research
- Find some opportunities to perform some novel research in your career

Contents of the Course



Chapters	Teacher	Class hours	Discus. hour	Total
Introduction	Qi Fei	8	4	12
Combustion Thermodynamics	Qi Fei			
Reaction kinetics and Pollutants	Qi Fei			
Conservation Equations for Reacting Flows	Li Lei	2	1	3
Detonations	Li Lei	2	1	3
Laminar Premixed Flame and Diffusion Flame	Li Lei	6	3	9
Flame-Flow Interaction	Li Lei	4	2	6
Droplet Evaporation and Combustion	Li Lei	4	2	6
Turbulent Flame	Li Lei	4	2	6
Discussion and Progress report				3

Working Group



Group	Name		
1	徐亮亮	杨晓媛	石天翔
2	刘 鹏	刘春江	李 伟
3	邹家标	戚荣杰	赵路甜
4	陈夏敏	傅 宸	陶 杰
5	李天宇	王思睿	

Homework, Exam and Scores



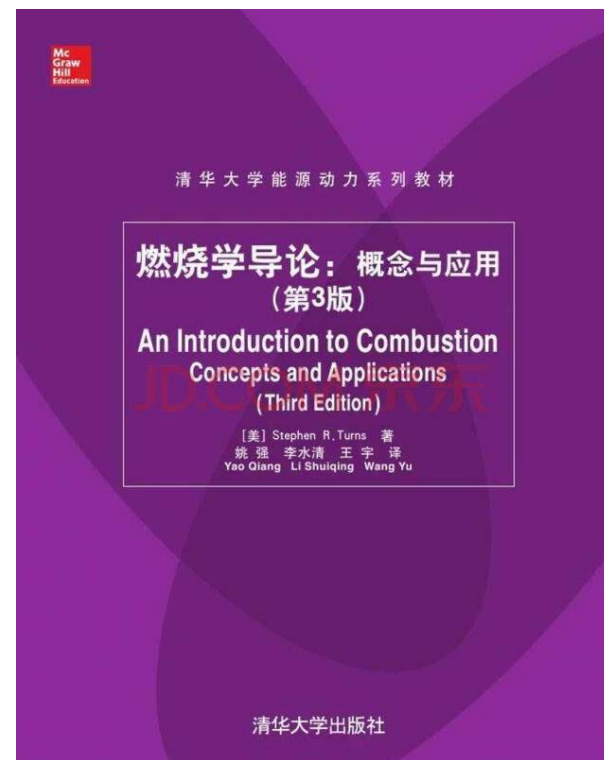
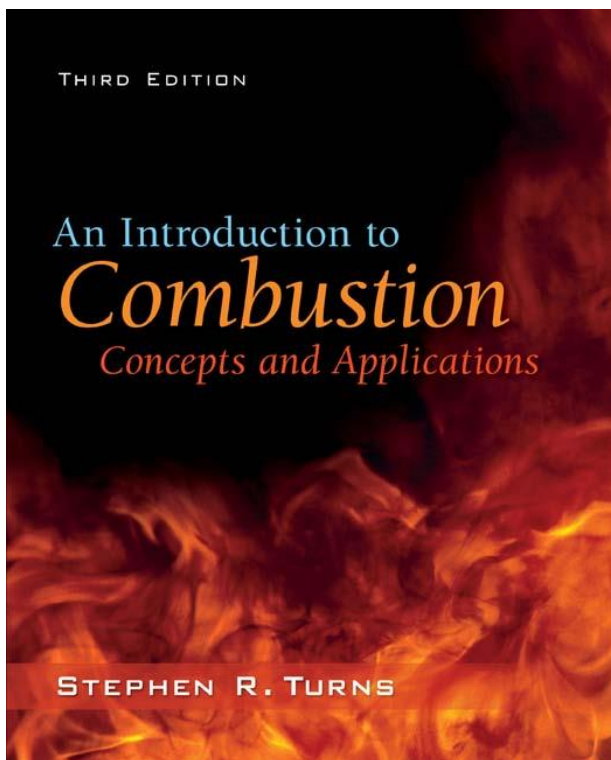
The **final scores** are weighted basing on your homework, presentation and the final report.

1. A homework will be assigned after every week's course.
2. Then the each working group works together to discuss and gives a solution.
3. One delegate of the each working group will present the homework (8-10 mins PPT + 5 mins Q&A) in the next course. All students rotate to make the presentation.

成绩：根据平时上课的PPT汇报以及最后的总结汇报

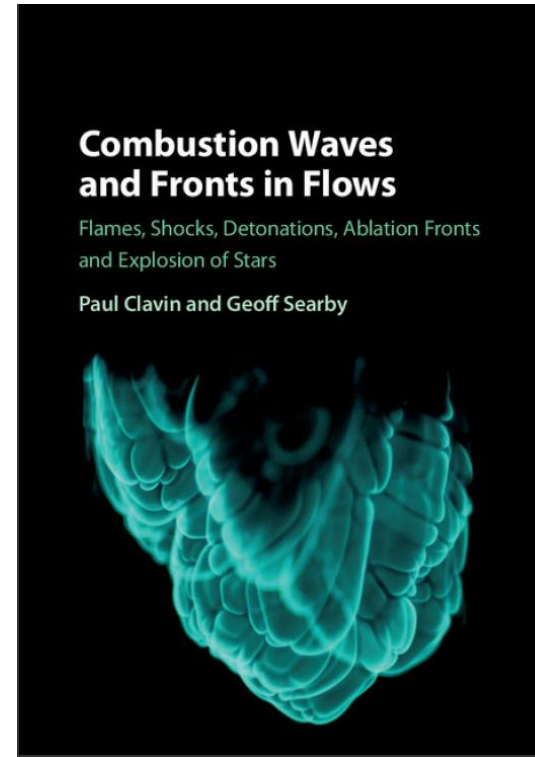
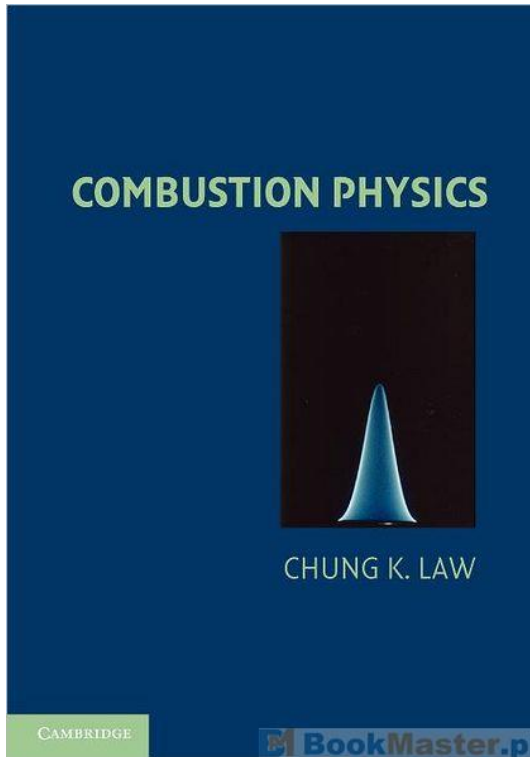
作业：每次课程之后会布置课后作业，小组成员一起讨论合作完成；下次上课时每组选出一位代表，用PPT汇报该小组的作业；汇报时间8-10分钟，讨论5分钟；所有学生都需要汇报。

References



- Stephen R. Turns, An Introduction to Combustion: Concepts and Applications, The Third Edition
- Yao Qiang et al. (姚强等人), 燃烧学导论：概念与应用，第三版

References



- Chung K. Law, Combustion Physics
- Paul Clavin and Geoff Searby, Combustion Waves and Fronts in Flows: Flames, Shocks, Detonations, Ablation Fronts and Explosion of Stars

Outline of Chapter 1



- **Combustion with human being**
- **Motivation to study combustion**
- **The history of combustion science and technology**
- **Definition of combustion**
- **Combustion research methods**
- **Introduction of Combustion Institute (CI) and publications**

Combustion and Civilization



- The control of fire by early humans was a turning point in the cultural aspect of human evolution.
- Fire provided a source of warmth, protection, and a method for cooking food.

Combustion and Civilization



Combustion influences the progress of human civilization

Pottery Age (B. C. 6000)



Furnace Temperature

800-1050 °C

陶器时期

Bronze Age (B.C. 3000)



Furnace Temperature

>1080 °C

青铜器时期

Iron Age (B.C. 1510)



Furnace Temperature

1500-2000 °C

铁器时期

Combustion and Civilization



Fire has been used in ancient war and affect the progress of history



Fire Bulls, Warring States Period
火牛阵-战国



Battle of Chibi
赤壁之战-三国



Battle of Yiling (夷陵之战-三国)



Battle of Poyang Lake (鄱阳湖水战-明)



Combustion in Daily Life





Combustion in Industry



Combustion in Mobility





Combustion in Environment and Climate



Combustion in Fire Disaster



Outline of Chapter 1

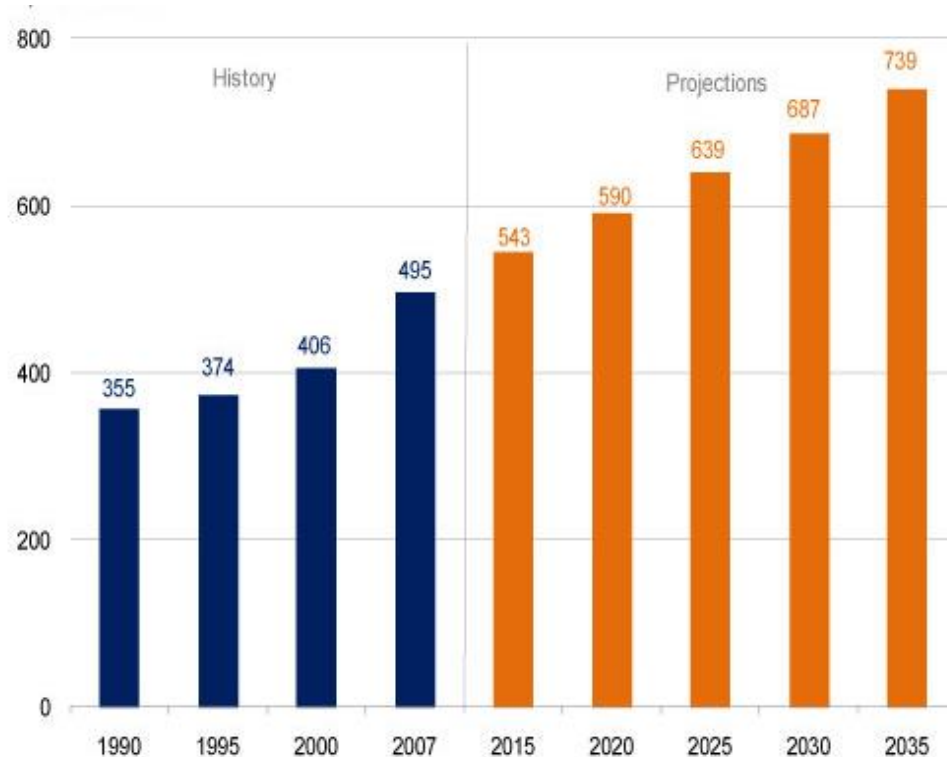


- **Combustion with human being**
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Combustion, Energy and Fuel

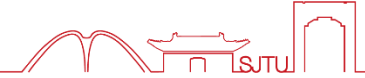


World energy consumption (quad. Btu)

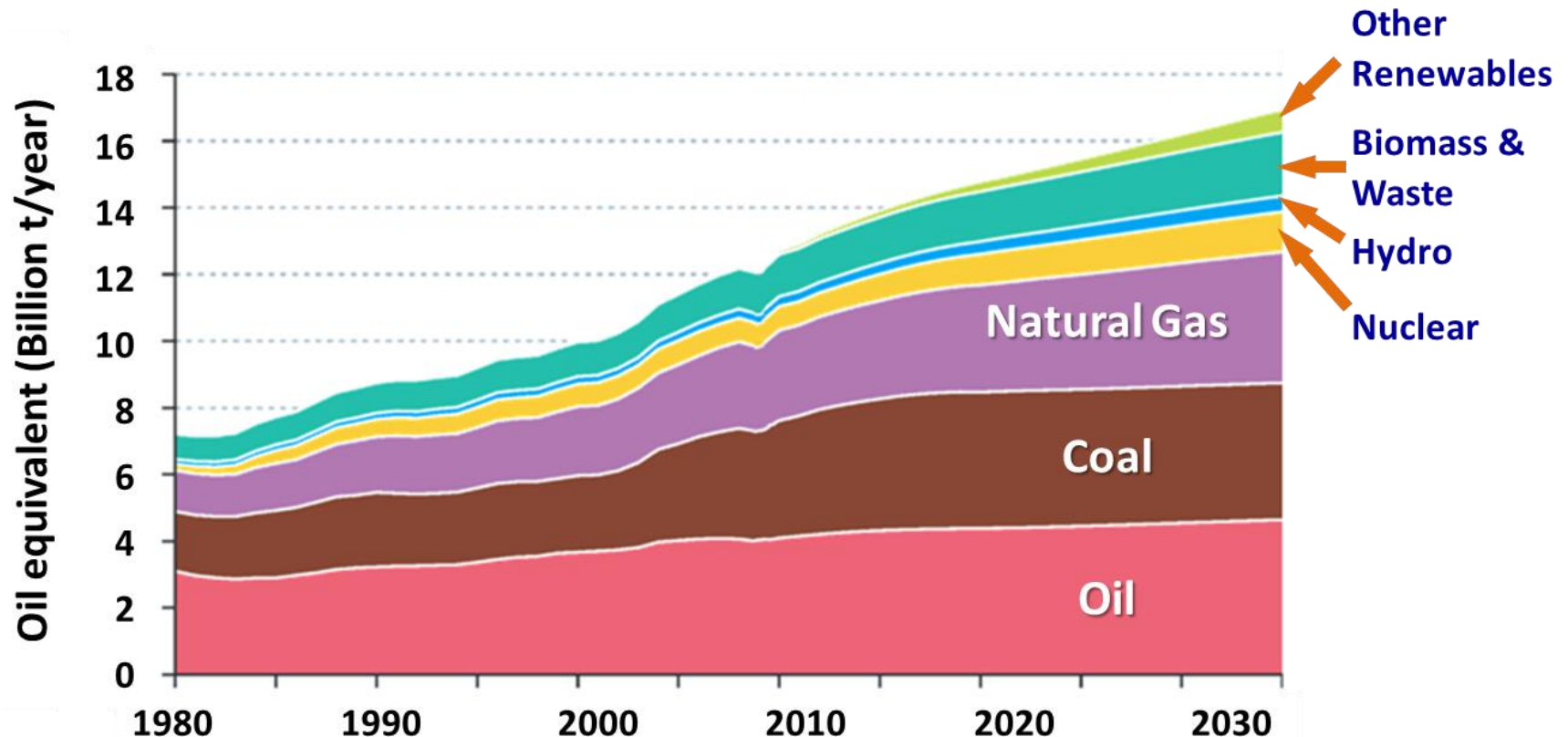


- Combustion provides >85% of global primary energy consumption.
- This key role will be maintained in the foreseeable future.

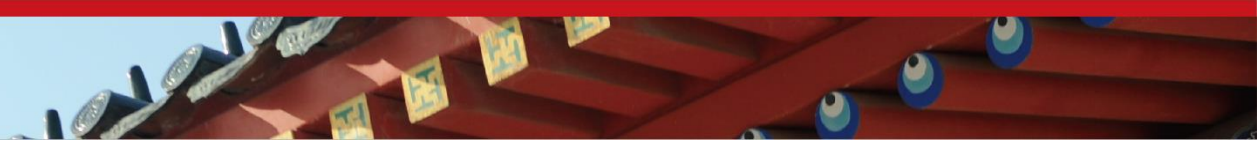
Combustion, Energy and Fuel



World energy by fuel type



- Most of energy sources comes from fossil fuels such as oil, coal, and natural gas.
- Renewable energy has been growing rapidly.



Combustion, Energy and Fuel



Status of world energy consumption

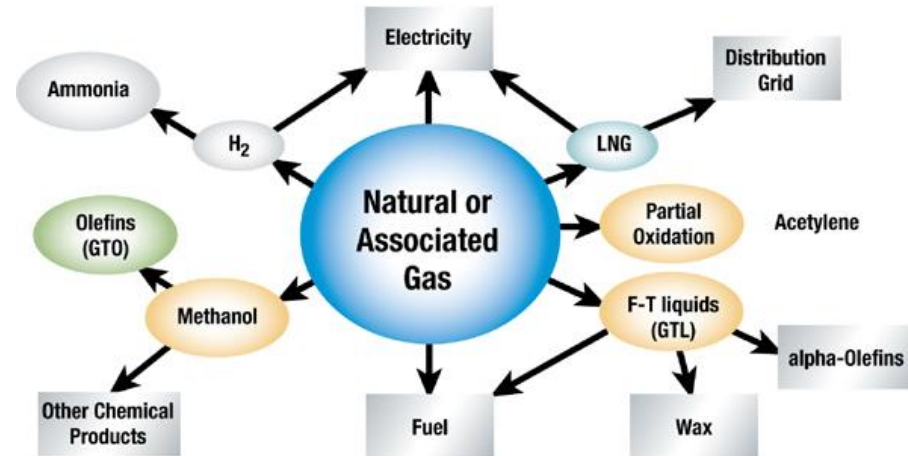
2014

Combustion, Energy and Fuel

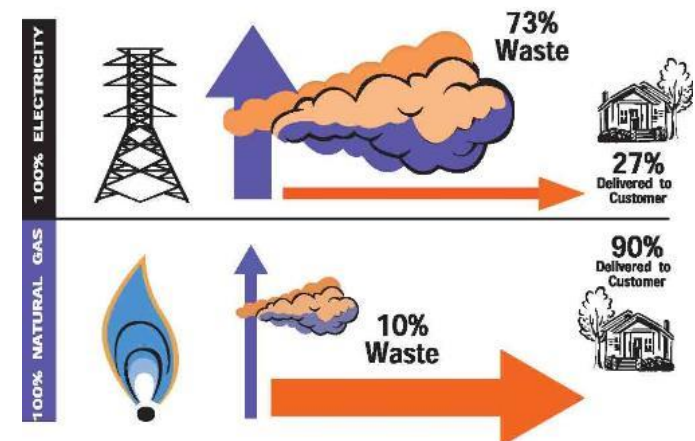
- **Natural gas** is a naturally occurring hydrocarbon gas mixture consisting primarily of methane, commonly including varying amounts of other higher alkanes, and sometimes a small percentage of carbon dioxide, nitrogen, hydrogen sulfide, or helium.



Natural Gas Utilization



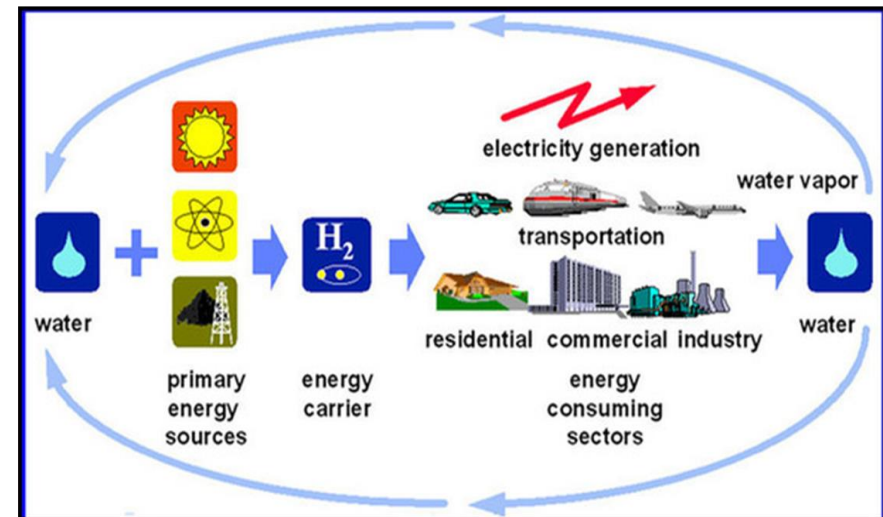
Natural Gas is Cleaner, More Efficient Than Coal-Fired Electric Power



Combustion, Energy and Fuel



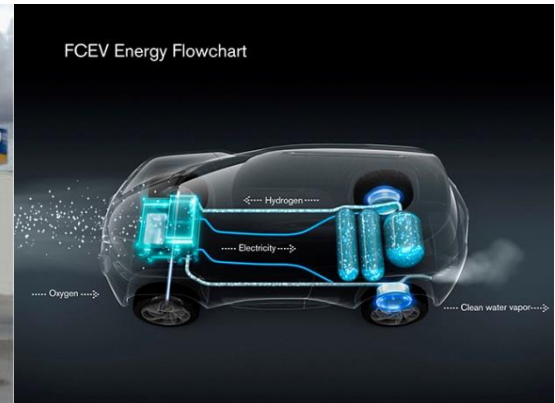
- **Hydrogen** can be considered as a clean energy carrier similar to electricity.
- Hydrogen can be produced from various resources such as renewable energy and nuclear energy.
- In the long-term, hydrogen will simultaneously reduce the emission of greenhouse gases & other pollutants.



Combustion, Energy and Fuel



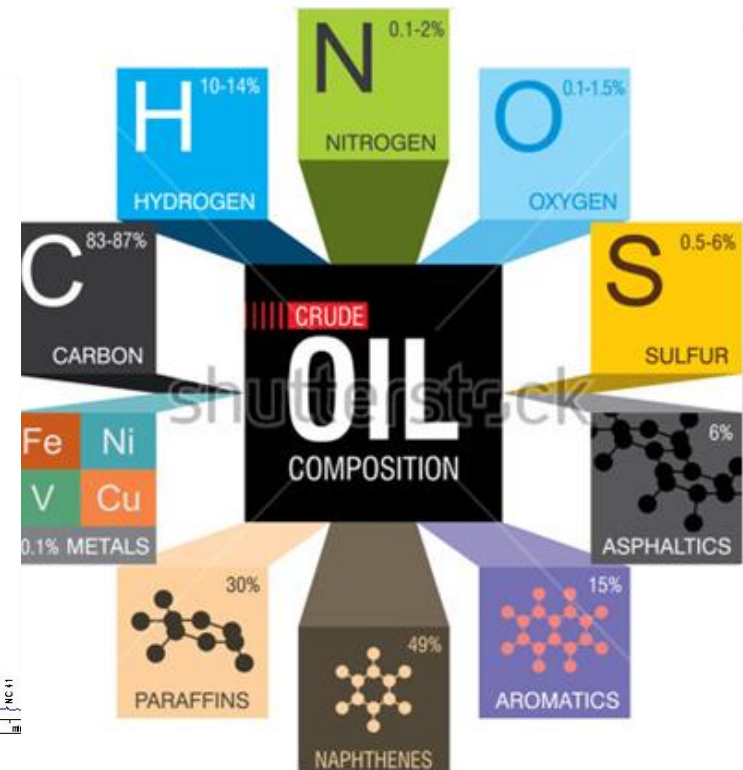
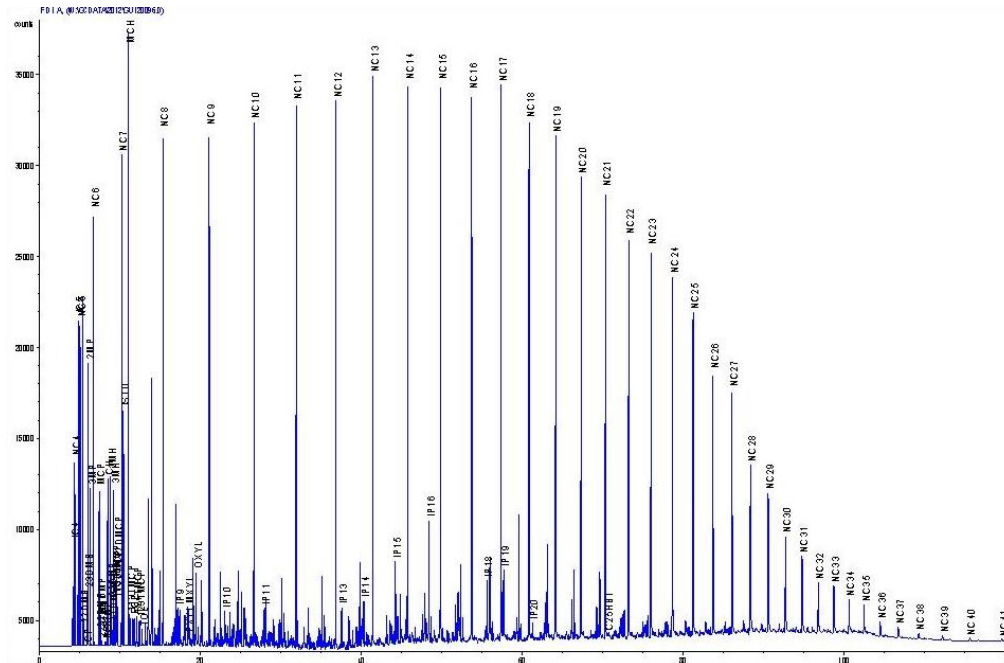
- Hydrogen fuel can provide motive power for liquid-propellant rockets, cars, boats and airplanes, portable fuel cell or stationary fuel cell.



Combustion, Energy and Fuel



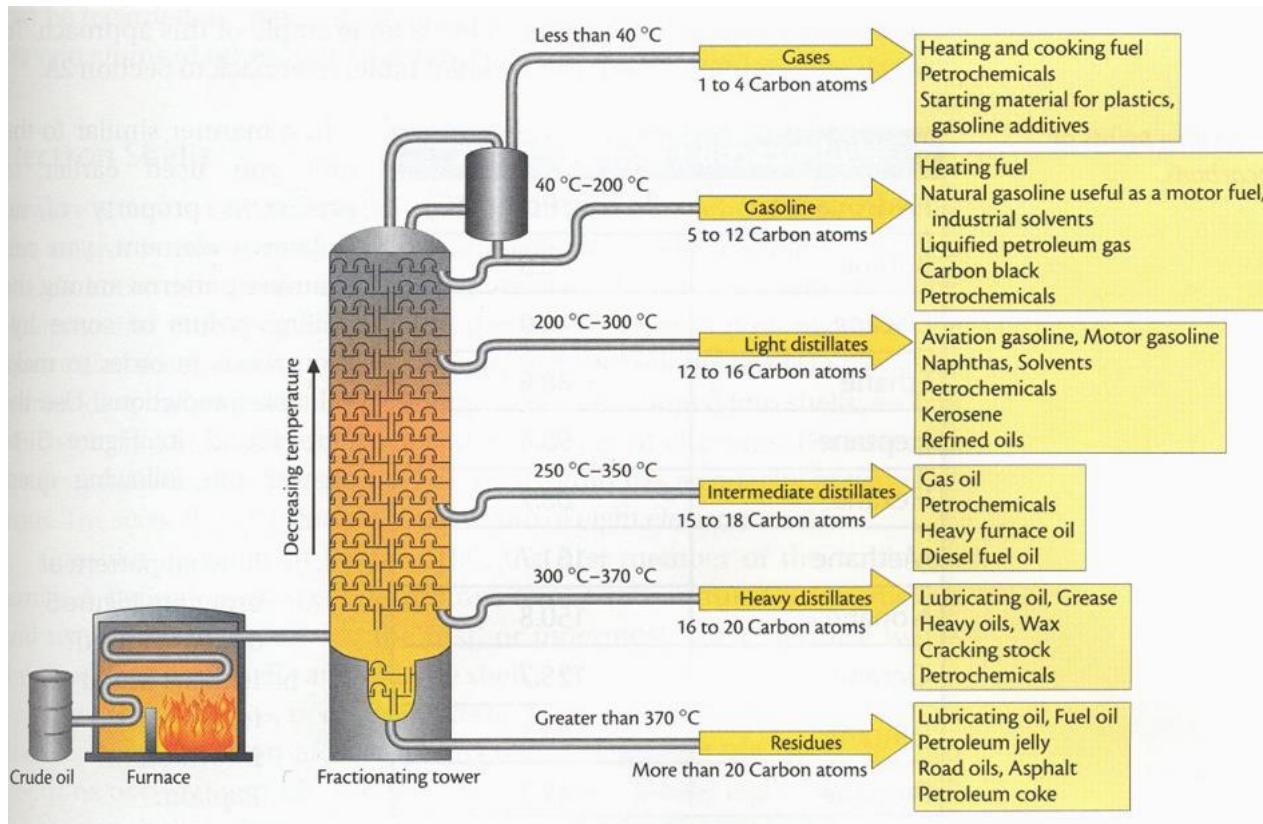
- **Petroleum** is a naturally occurring, yellow-to-black liquid found in geological formations beneath the Earth's surface, It consists of hydrocarbons of various molecular weights and other organic compounds.



Combustion, Energy and Fuel



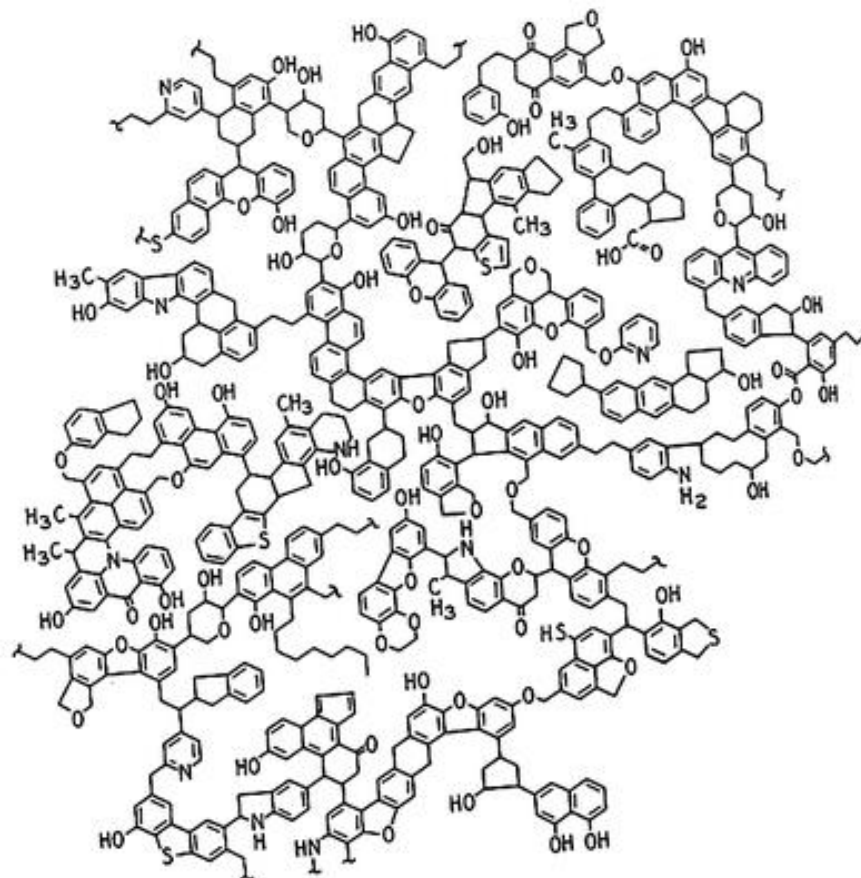
- Refining of petroleum produces more useful products such as petroleum naphtha, gasoline, diesel fuel, asphalt base, heating oil, kerosene, and liquefied petroleum gas.



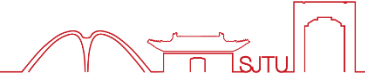
Combustion, Energy and Fuel



- **Coal** is one of the world's most important and abundant energy sources.
- Coal consists of thousands of organic compounds that are mainly hydroxyl- and alkyl-substituted aromatic and heteroaromatic rings.



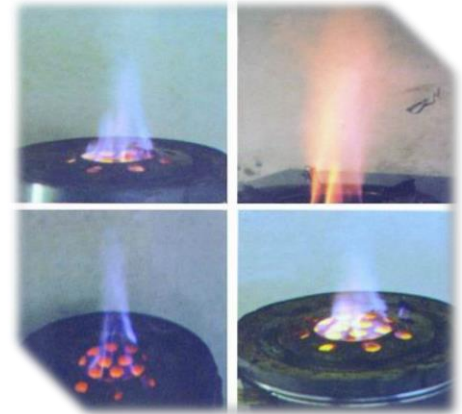
Combustion, Energy and Fuel



Utilization of Coal



steam engine



daily life



electricity generation



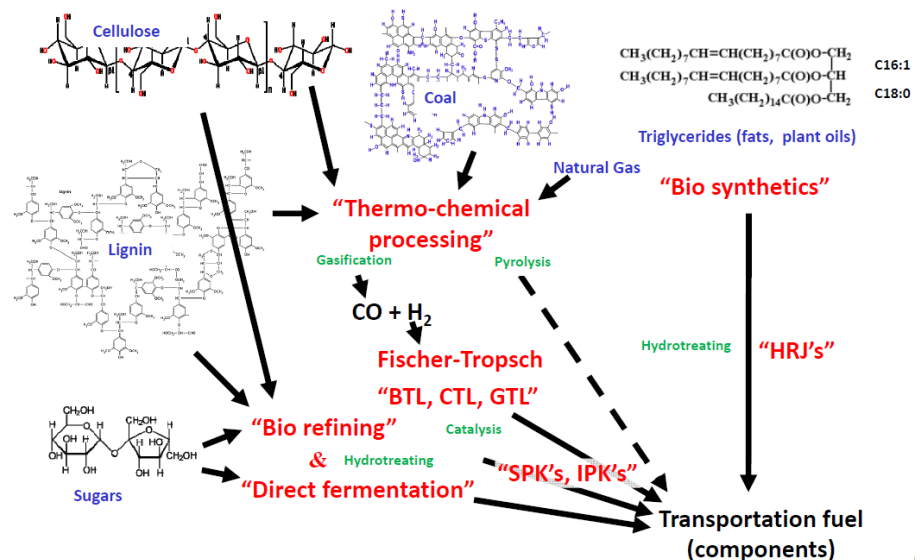
boiler



metallurgy

Combustion, Energy and Fuel

Biofuel is a fuel that is produced through contemporary biological processes, such as agriculture and anaerobic (无氧的) digestion, rather than a fuel produced by geological processes such as those involved in the formation of fossil fuels, such as coal and petroleum, from prehistoric biological matter. Biofuels can be derived directly from plants, or indirectly from agricultural, commercial, domestic, and/or industrial wastes.

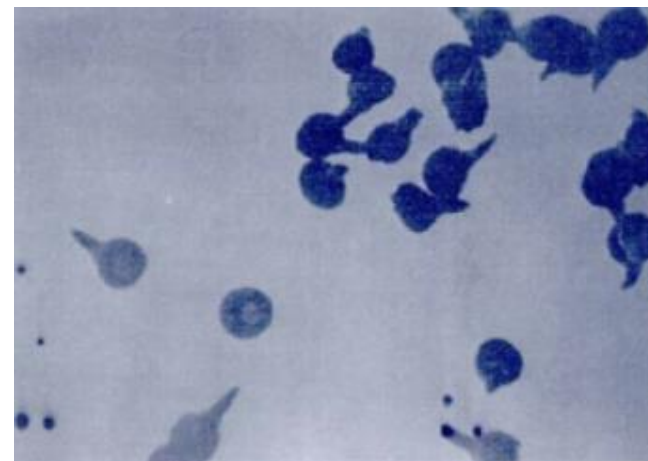
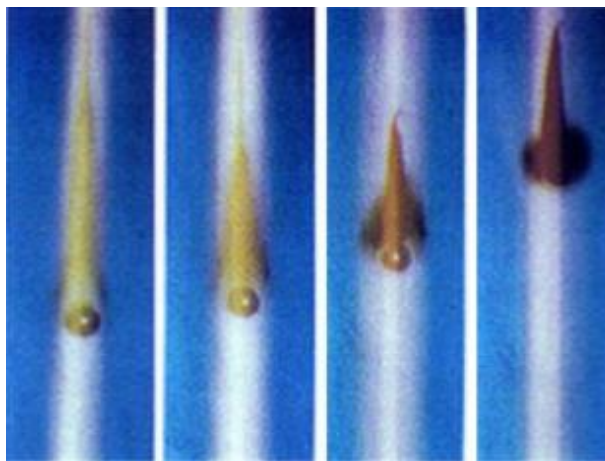
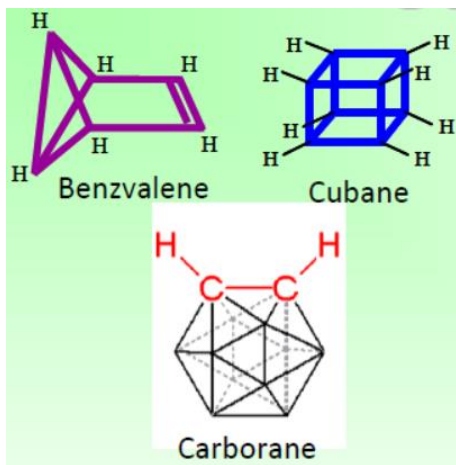


Combustion, Energy and Fuel



Energetic Fuels

□ Three types of energetic fuels



Strained molecules

Metallized propellants

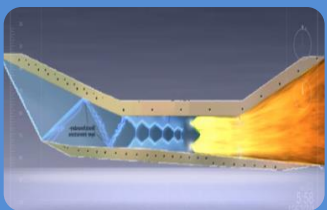
Nanoparticles

□ High energy density and burning rate

□ New concepts in propellant formulation

Combustion and Power



IC engine	Gas turbine	Aircraft engine	Scramjet engine	Rocket engine
				
				
<u>Emissions</u>	<u>Compression Ratio</u>	<u>Thrust-Weight Ratio</u>	<u>Mach Number</u>	<u>Actual Load</u>
EU VI	24-30	10-13	7-10	100-130 ton
China 5	<15	6-8	5-6	8-10 ton
Clean combustion	High density combustion	High temperature rise combustion	Supersonic combustion	Stable combustion

High performance, high efficiency and low emissions combustion

Combustion and Power



- In an **Internal Combustion (IC)** engine, the ignition and combustion of the fuel occur within the engine itself. The engine then partially converts the energy from the combustion to work.

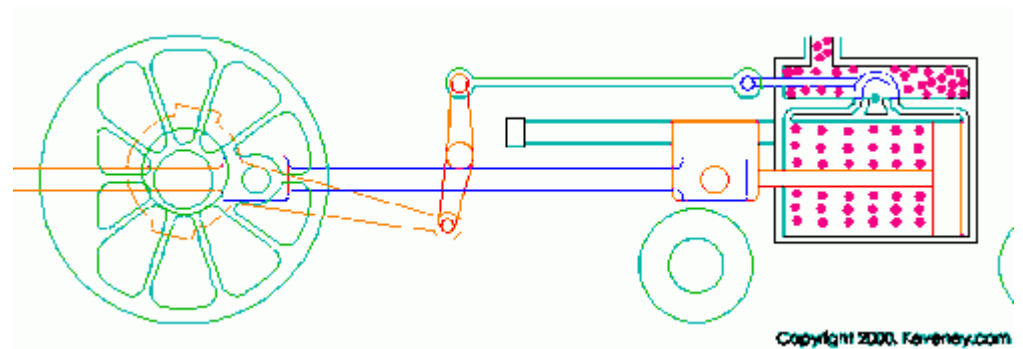
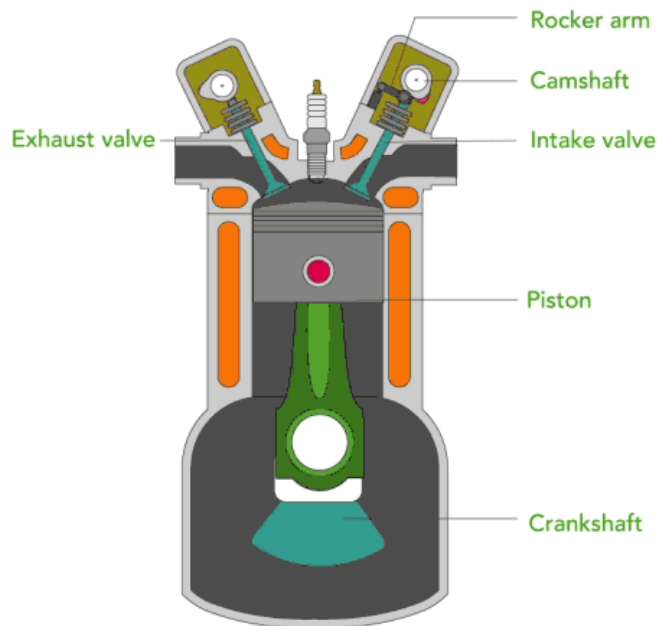


Internal Combustion (IC) Engine

Combustion and Power



Internal Combustion Engine

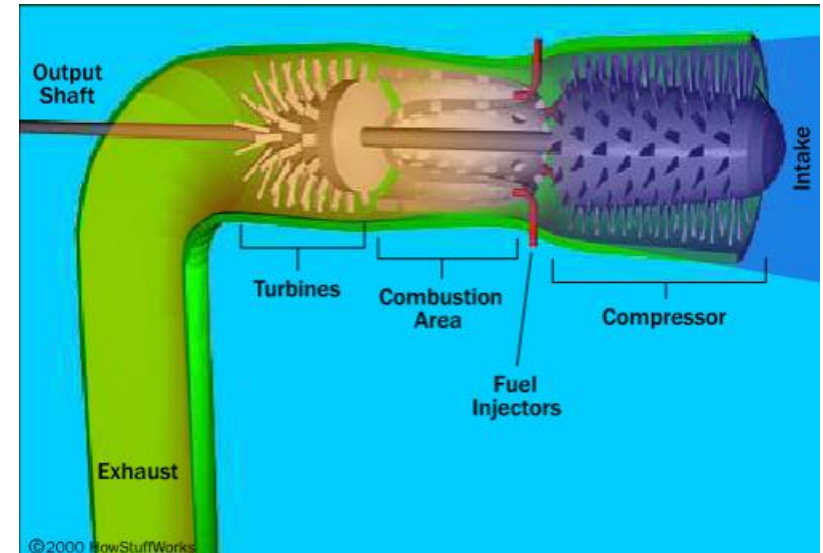
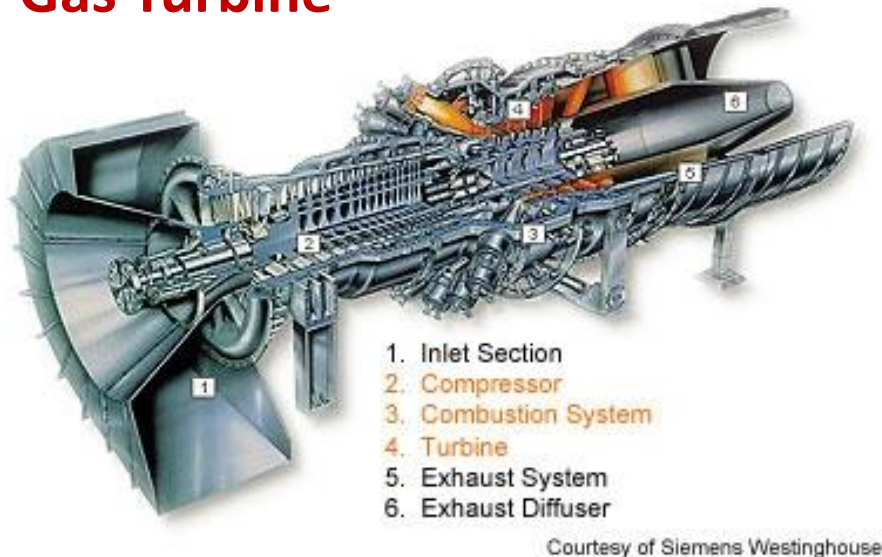


- The engine consists of a fixed cylinder and a moving piston. The expanding combustion gases push the piston, which in turn rotates the crankshaft. Ultimately, through a system of gears in the powertrain, this motion drives the vehicle's wheels.

Combustion and Power



Gas Turbine

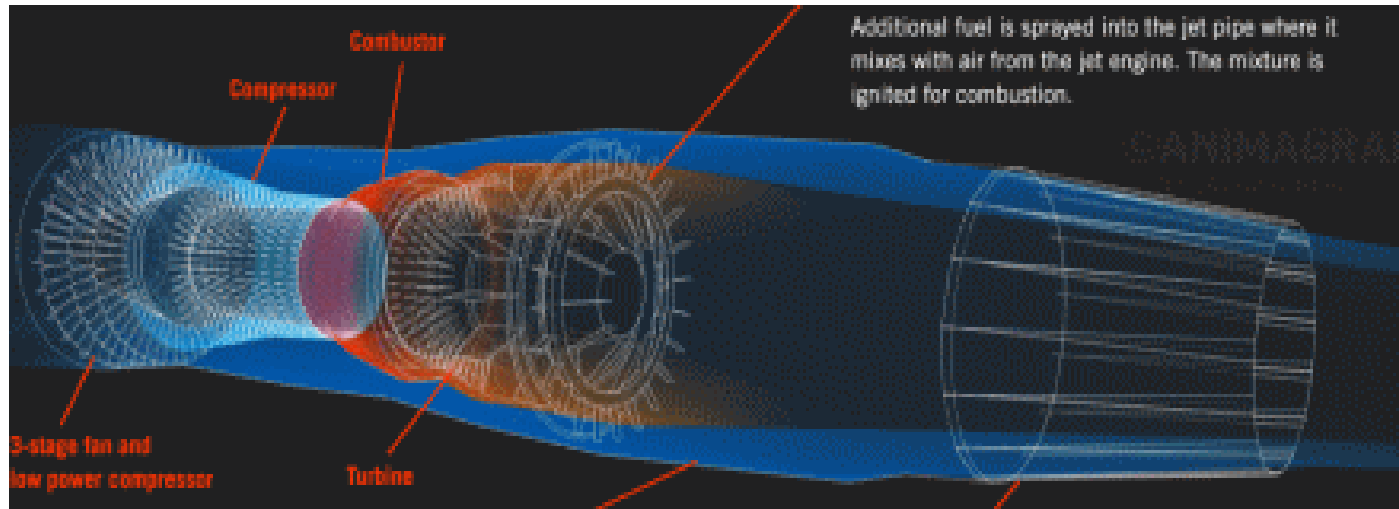


- The **gas turbines** being installed in many of today's **natural-gas**-fueled power plants are complex machines.
- The combustion produces a high temperature, high pressure gas stream that enters and expands through the turbine section, where it expands down to the exhaust pressure, producing a shaft work output in the process.

Combustion and Power



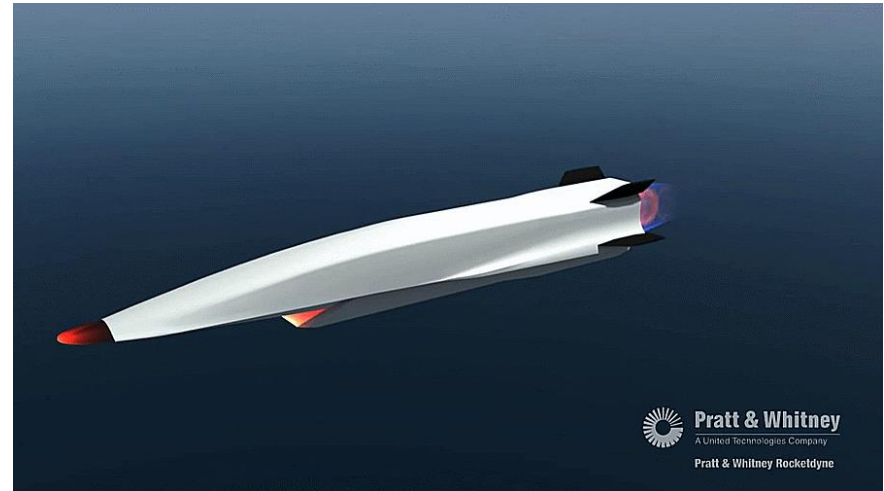
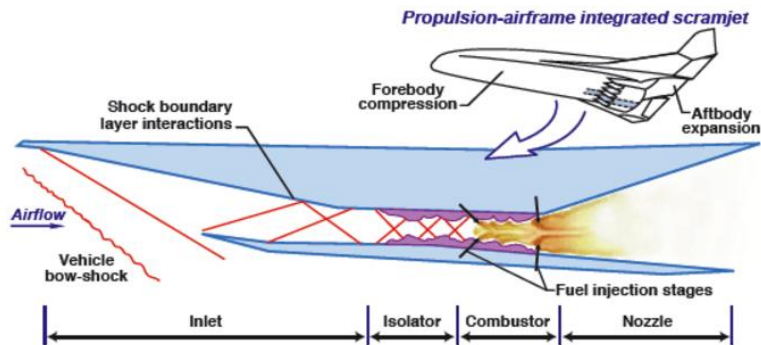
Jet engine



- A **jet engine** is a machine that converts **energy-rich, liquid fuel** into a powerful pushing force called thrust. The thrust from one or more engines pushes a plane forward, forcing air past its scientifically shaped wings to create an upward force called lift that powers it into the sky.

Combustion and Power

Scramjet engine

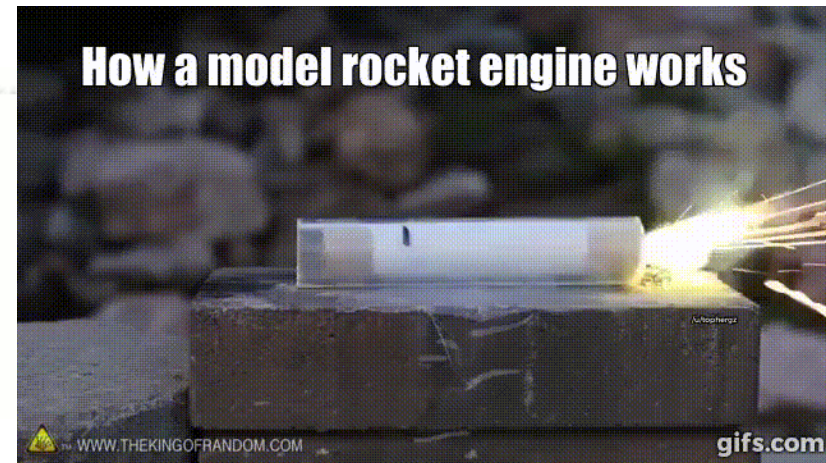
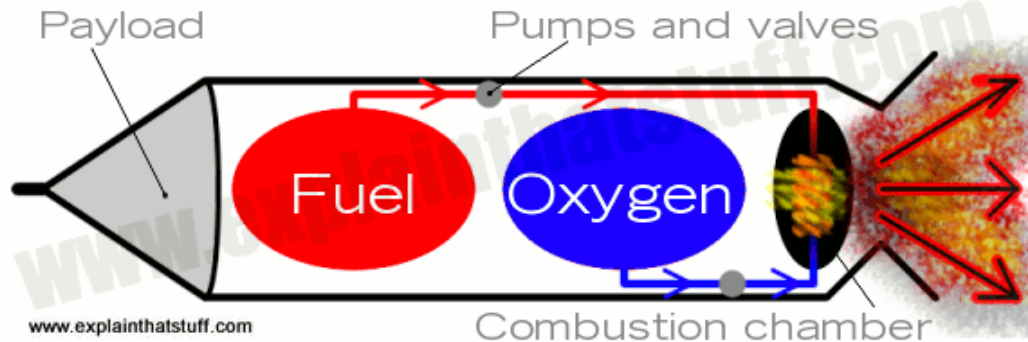


- A **scramjet** operates by combustion of fuel in a stream of air compressed by the forward speed of the aircraft itself, as opposed to a normal jet engine, in which the compressor section (the fan blades) compresses the air. The airflow through the engine remains supersonic, or greater than the speed of sound. Scramjet powered *vehicles* are envisioned to operate at speeds up to at least Mach 7.

Combustion and Power



Rocket engine

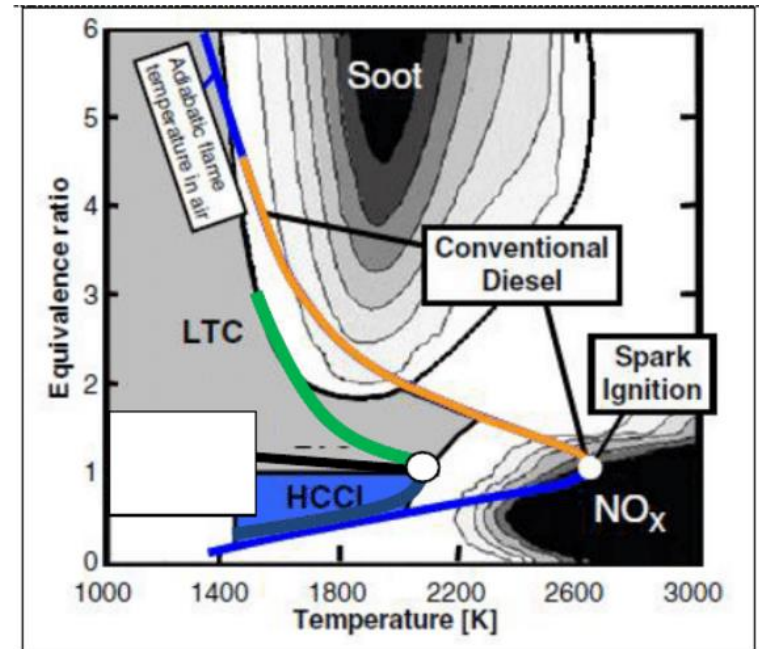
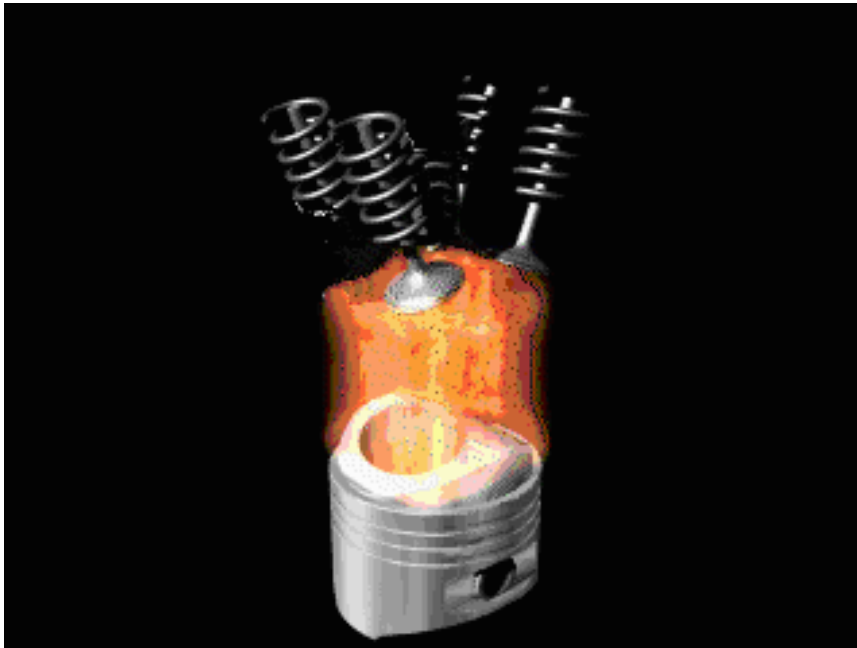


- In a **rocket engine**, fuel and a source of oxygen, called an oxidizer, are mixed and exploded in a combustion chamber. The combustion produces hot exhaust which is passed through a nozzle to accelerate the flow and produce thrust.
- There are two main categories of rocket engines; liquid rockets and solid rockets

Combustion and Power

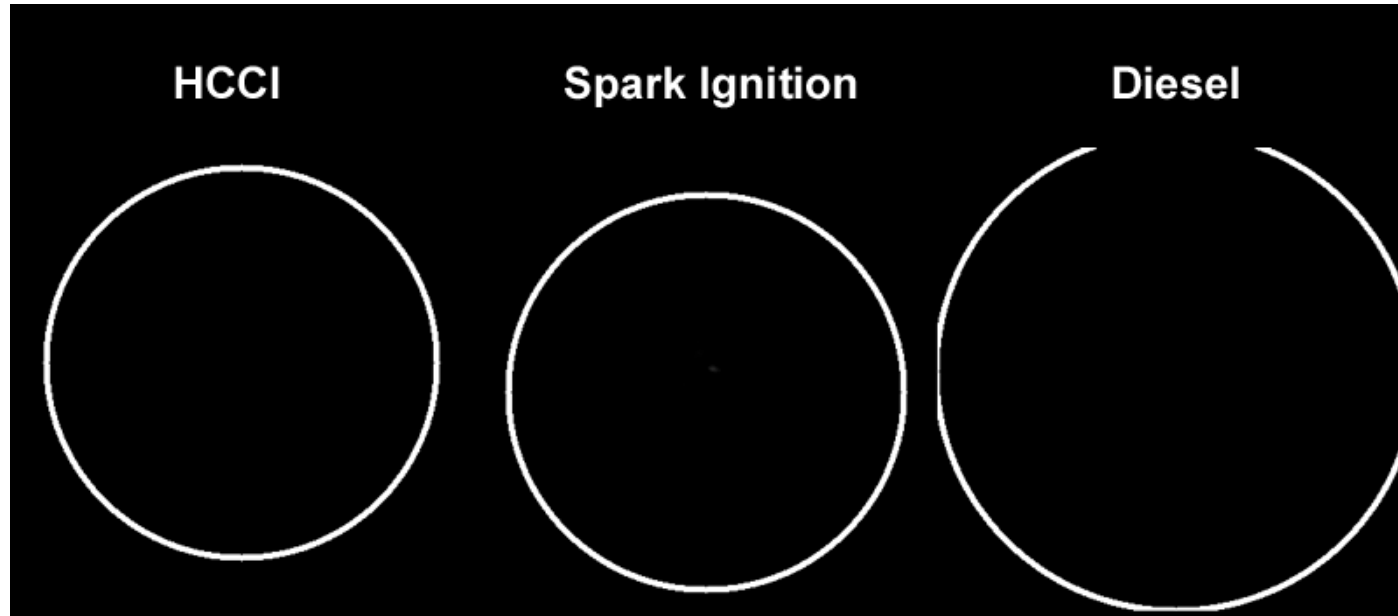


Advanced engines with high efficiency and low pollutant emissions



- Due to the use of Low Temperature Combustion (LTC) mode, **HCCI** (Homogeneous Charge Compression Ignition) engines achieve extremely low levels of nitrogen oxide (NO_x) emissions.

Combustion and Power

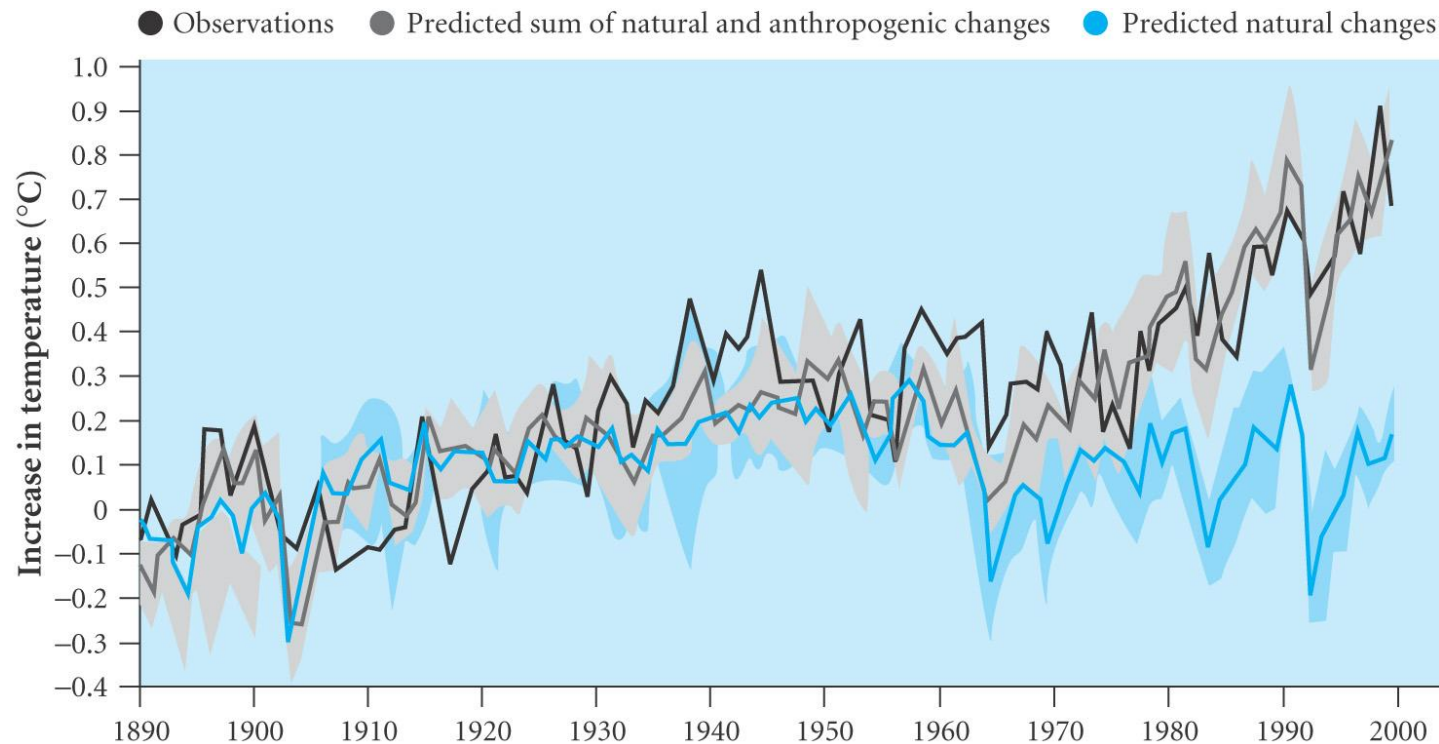


- **HCCI** is a hybrid combustion strategy that is being actively researched. Like a spark-ignition engine, a mixture of fuel and air is inducted into the chamber; but instead of igniting the mixture with a spark plug, the high temperature that results from compression of the mixture to a high pressure causes the mixture to spontaneously react.

Combustion and Environment

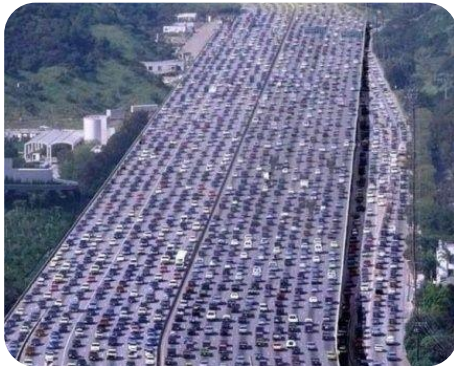
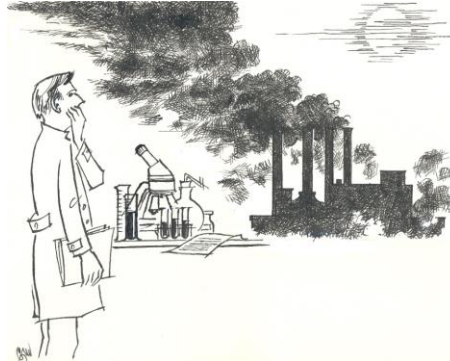


Global temperature increase



- Combustion affects global warming by dumping lots of carbon dioxide (CO_2) and other combustion byproducts (CH_4 , NO_x and so on) into the atmosphere.

Combustion and Environment



Air pollutant emissions from vehicles in China (Year 2013)

Pollutant	Emissions (Mton)
NOx	6.406
HC	4.312
CO	3.440
PM	0.594
Total	45.709

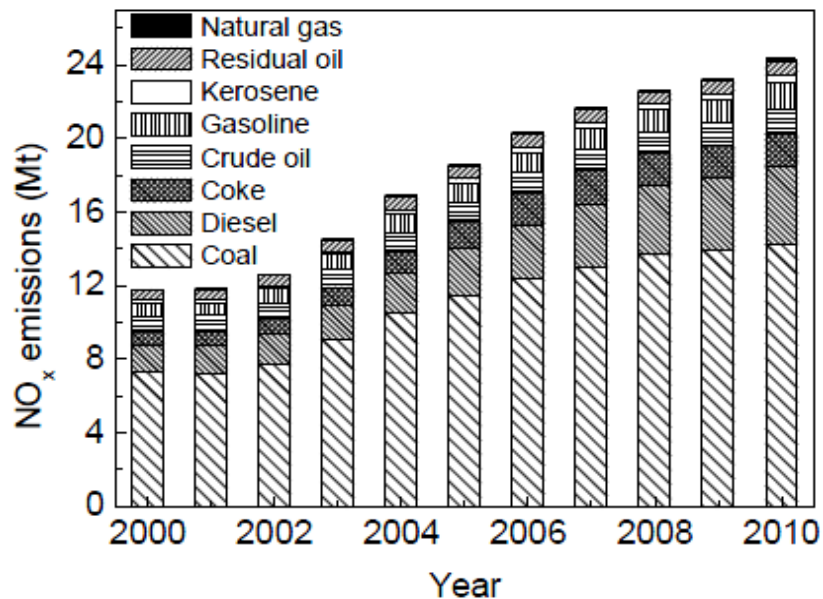
**Combustion produces a great amount of air pollutants,
such as NOx, SOx, VOCs, PAHs, Soot**

Combustion and Environment

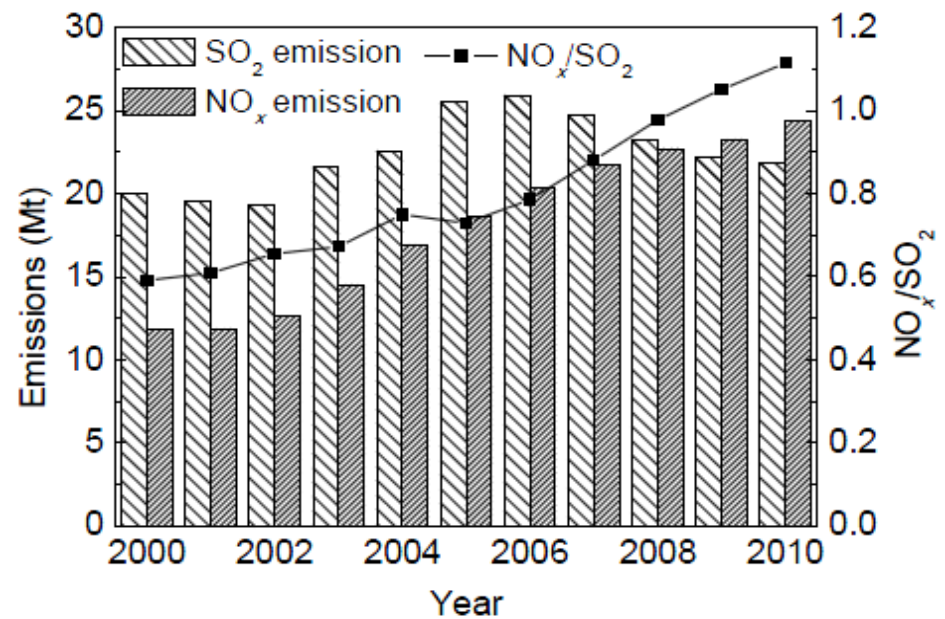


Combustion is the main emission source for NO_x and SO_x

NO_x emission from fossil fuel combustion in China



Comparison of the trends of NO_x and SO_x emissions



Combustion and Environment



- The combustion of fossil fuel and biomass produces soot particle, which is one of major source of haze.

Combustion and Environment



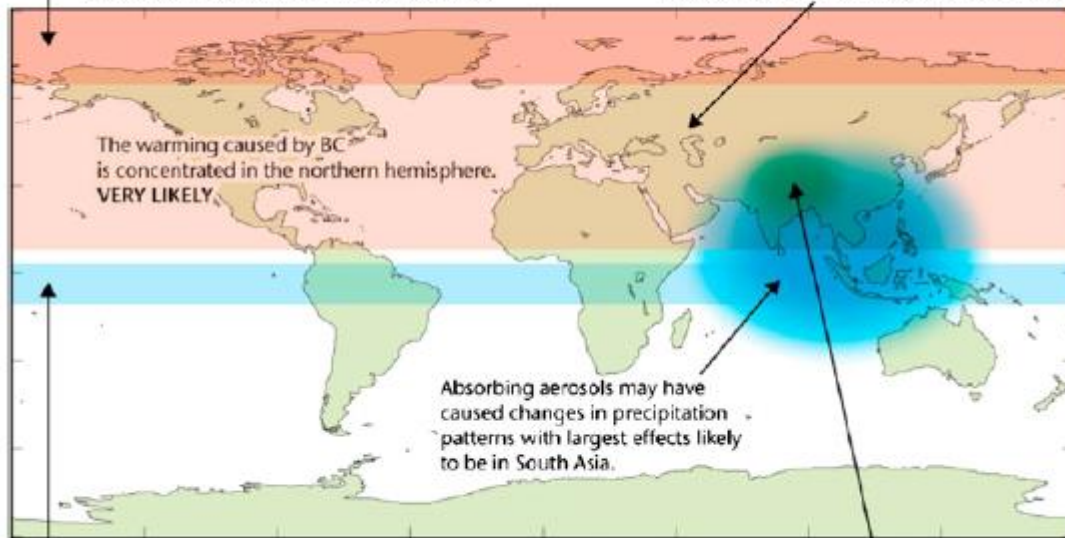
- Combustion is the main emission source for soot (black carbon).
- Soot is known to be carcinogenic (致癌的) and mutagenic (致畸的) and is a major contributor to particulate matter 2.5 (PM2.5).

Impact environment

Climate effects of black carbon emissions

The impact of BC on snow and ice causes additional warming in the Arctic region and contributes to snow/ice melting. **VERY LIKELY BUT MAGNITUDE UNCERTAIN**

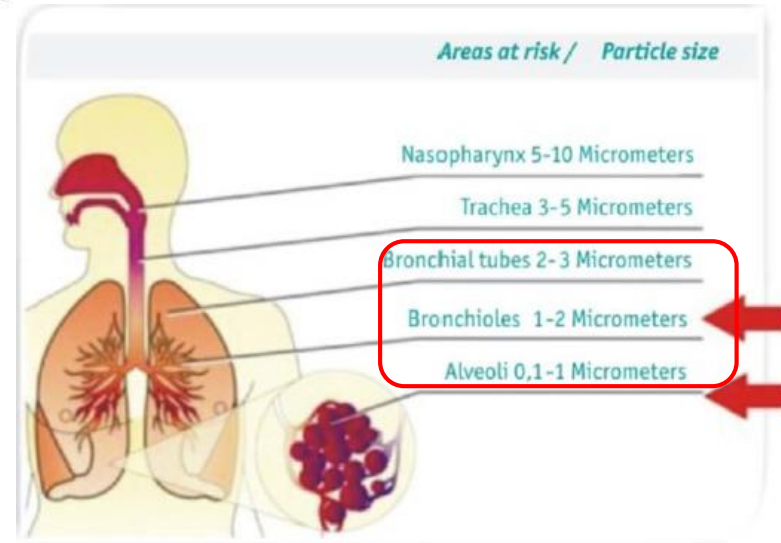
BC in northern hemisphere mid-latitude snow leads to earlier springtime melt and reduces snow cover in some regions. **LIKELY BUT MAGNITUDE UNCERTAIN**



The hemispheric nature of the BC forcing causes a northward shift in the ITCZ. **LIKELY.**

Absorbing aerosols may cause circulation changes over the Tibetan Plateau and darkening of the snow. The importance of this for glacier melting is unknown.

Impact human health



Combustion and Fire Safety



Forest and city fire

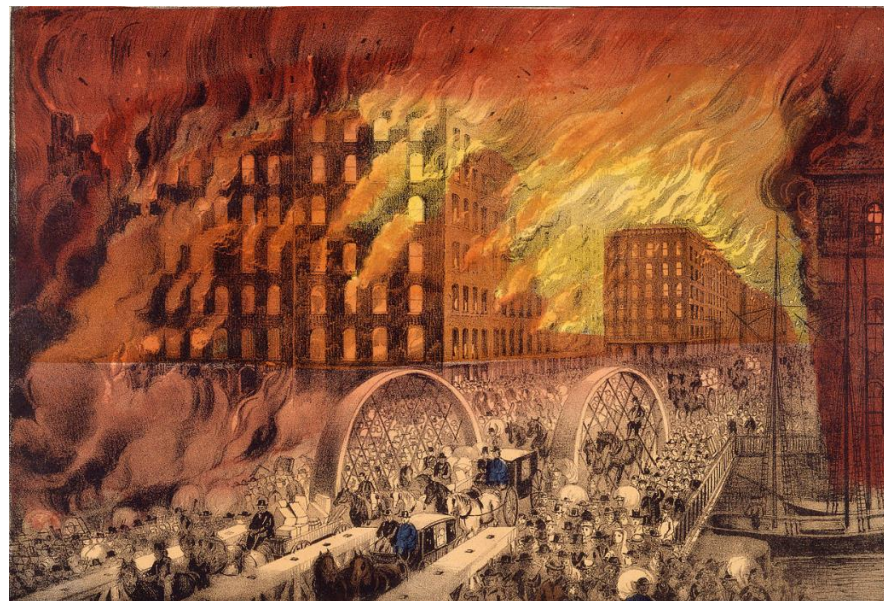


- The 1987 Daxing'anling Wildfire (大兴安岭森林火灾), was a major wildfire. The burning lasted almost a month, The fire covered about 10,000 square kilometres, of which 6500 square kilometres was forestry.

Combustion and Fire Safety



Forest and city fire



- The Great Chicago Fire was a conflagration that burned from October 8, to October 10, 1871. The fire killed up to 300 people, destroyed roughly 3.3 square miles (9 km²) of Chicago, Illinois, and left more than 100,000 residents homeless.

Combustion and Fire Safety

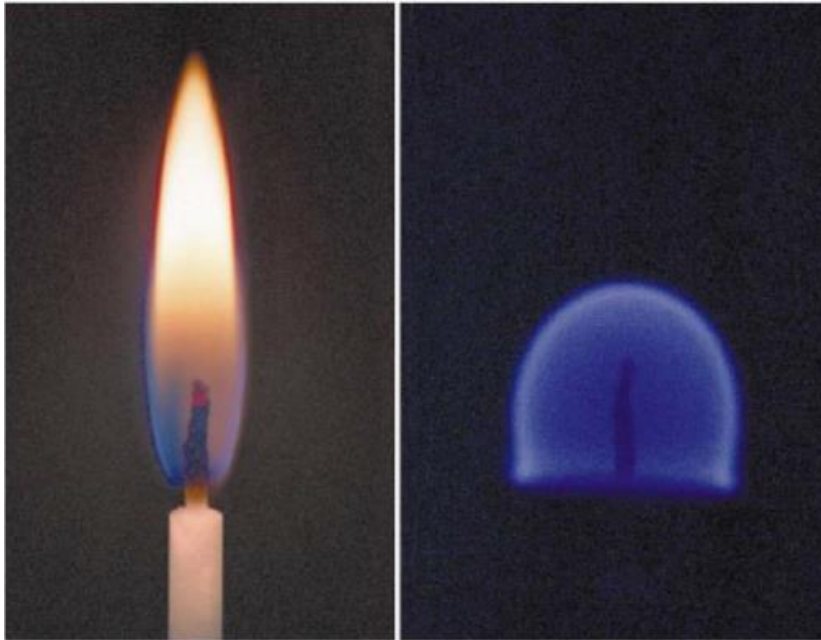


Fire whirl (火旋风)



- Fire whirl is a whirlwind induced by a fire and often made up of flame or ash. Fire whirls may occur when intense rising heat and turbulent wind conditions combine to form whirling eddies of air. These eddies can contract into a tornado-like structure that sucks in burning debris and combustible gases.

Combustion and Aerospace

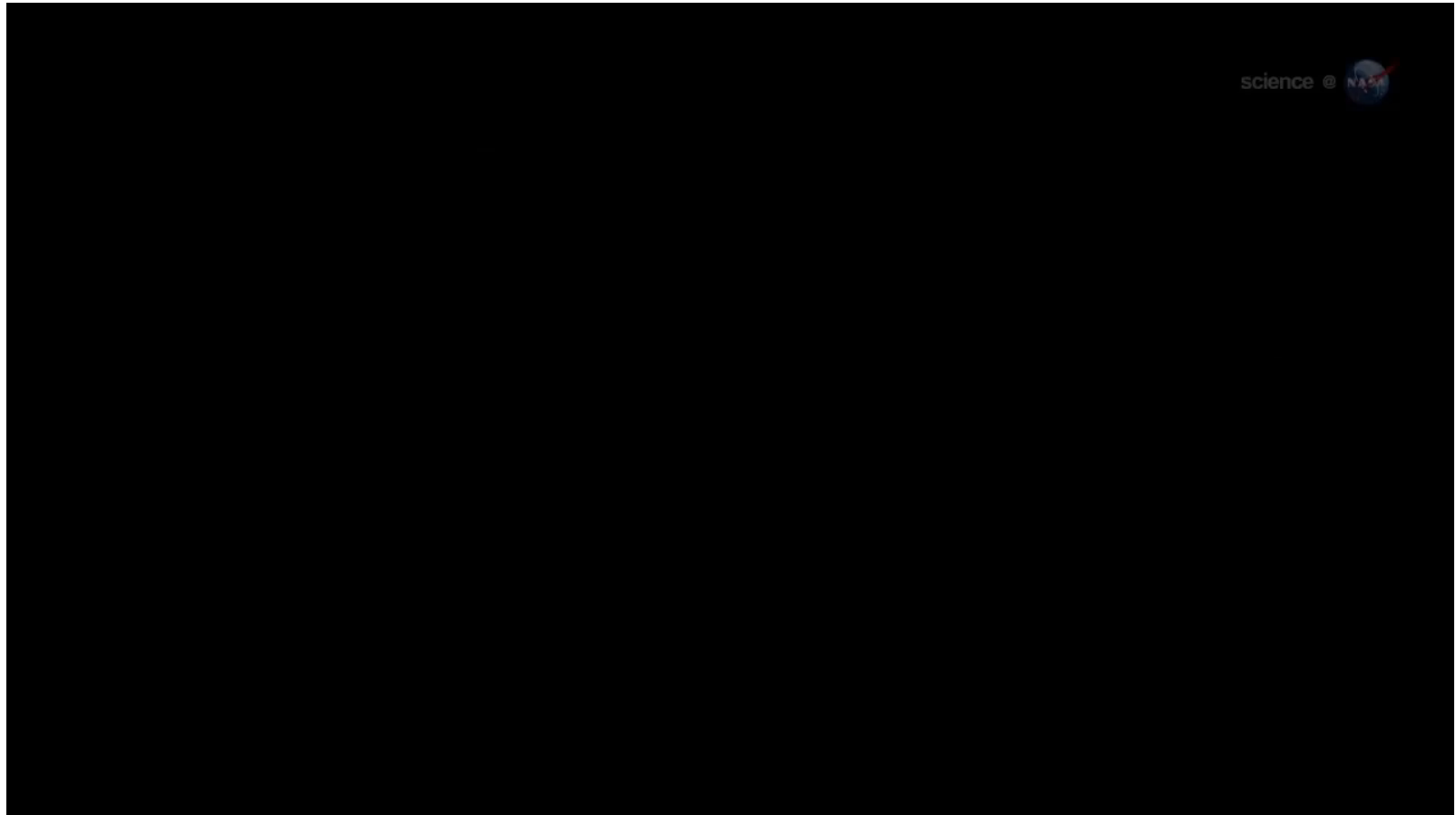


Microgravity produces a round, cooler candle flame with no soot. Primarily because microgravity provides an environment that lacks buoyant convection, which normally plays an important role in maintaining and shaping a flame on earth.

Combustion and Aerospace



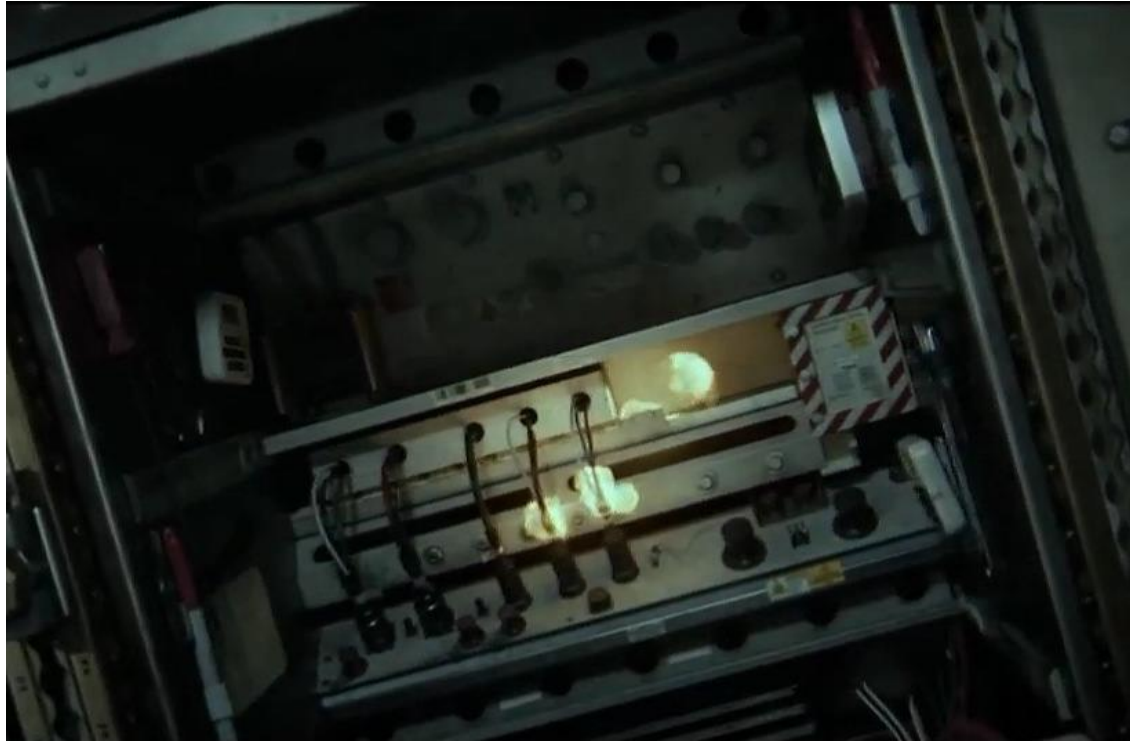
Fire aboard the International Space Station



Combustion and Aerospace

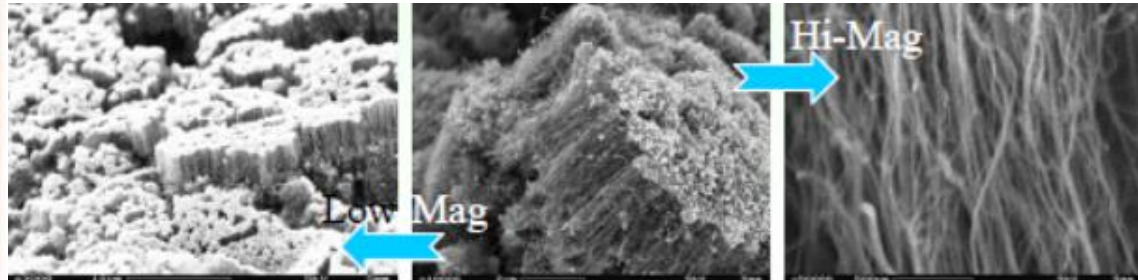
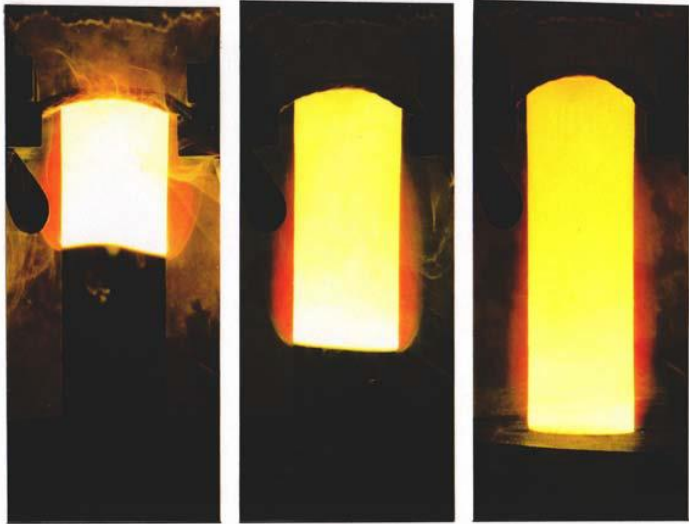


Fire aboard the International Space Station



Combustion studies in microgravity are important to spacecraft safety, it is essential to understand how fires are ignited in microgravity and how they spread.

Combustion and Materials Synthesis



Advantages of flame synthesis:

- ❑ High production rate & purity
- ❑ Nano-sized particles; increased catalyticity
- ❑ Functional materials

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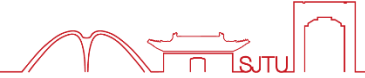
Human Using Fire



Human has come a long way from using fire to realizing fire!



Ancient Use and Thinking of Combustion



Ancient element concepts

Ancient China

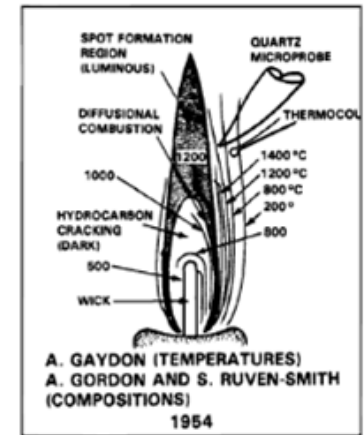
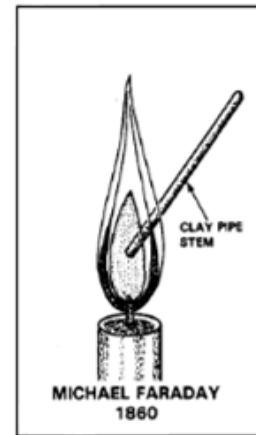
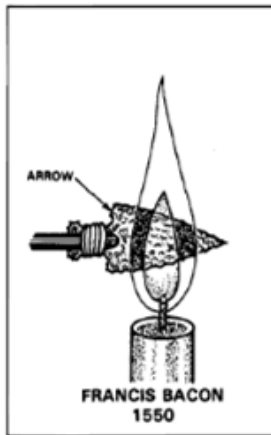


Ancient Greece



- Fire is one of the classical elements with features hot and dry
- The classical elements can be transformed by changing their features

Time Line of Combustion Research



1550 1600 1650 1700 1750 1800 1850 1900 1950

Phlogiston Theory
J.B. van Helmont, J. Becher

"The Chemical History of a Candle"
M. Faraday

Combustion as Oxidation
A. Lavoisier, P. S. Laplace

Chain reaction and thermal ignition (Nobel 1956)
Semenov, Hinshelwood

→ Theory of gas kinetics, free radicals, elementary reactions and reaction mechanisms for combustion applications

Phlogiston Theory (燃素说)

□ 1667, J. J. Becher, Germany

□ 1703, G. E. Stahl, Germany

Substances contain phlogiston & dephlogisticated air when burned

charcoal (+ Π) + air = fixed air(CO_2) + phlogisticated air
metal (+ Π) + air = calx (pure base) + phlogisticated air
charcoal (+ Π) + calx = metal (+ Π) + fixed air
metal (+ Π) + water = calx + inflammable air (Π)
metal (+ Π) + acid = salt + inflammable air (Π)
mercury (+ Π) + dephlogisticated air = calx of mercury
inflammable air (+ Π) + dephlogisticated air = water

J. J. Becher
(1635-1682)



G. E. Stahl
(1659-1734)



Challenges

- What is the principle of phlogiston?
- Why some substances gain mass when they burned?
- Why the volume of air is reduced when combustion?



Oxidation Theory of Combustion (氧化说)



□ 1777, A. Lavoisier, France

- The gain in weight by combustion is the result of combination with a large quantity of oxygen.
- The mass of the reactants equals the mass of the products.

A. Lavoisier
(1743-1794)



*Father of
modern chemistry*

J. Priestley
(1733-1804)



*Discoverer of
oxygen (1774)*



Nothing is lost, nothing is
created, everything is
transformed

~ Antoine Lavoisier



Beginning of combustion science and modern chemistry

Oxidation Theory of Combustion (氧化说)



The experiments of Faraday

The Chemical History of a Candle (1848)

Lecture I A Candle: The Flame—Its Sources—Structure—Mobility—Brightness

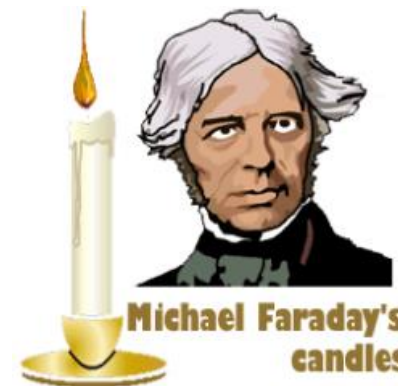
Lecture II A Candle: Brightness of the Flames—Air Necessary for Combustion—Production of Water

Lecture III Products: Water from the Combustion—Nature of Water—A Compound—Hydrogen

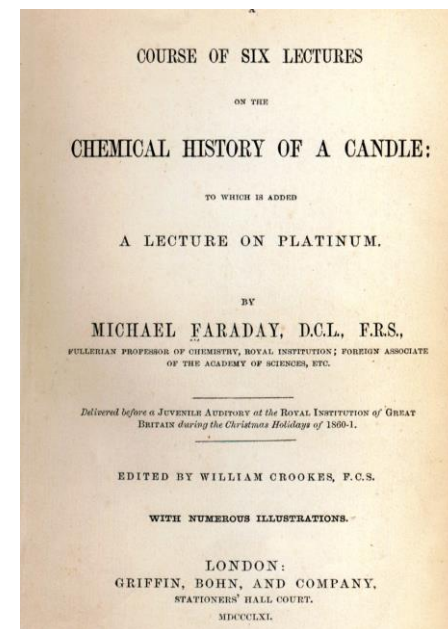
Lecture IV Hydrogen in the Candle—Burns into Water—The Other Part of Water—Oxygen

Lecture V Oxygen Present in the Air—Nature of the Atmosphere—Its properties—Other Products from the Candle—Carbonic Acid—Its Properties

Lecture VI Carbon or Charcoal—Coal-Gas—Respiration and its Analogy to the Burning of a Candle—Conclusion



M. Faraday
(1791-1867)



Development of Thermodynamics



Thermodynamics was founded by:



James Watt



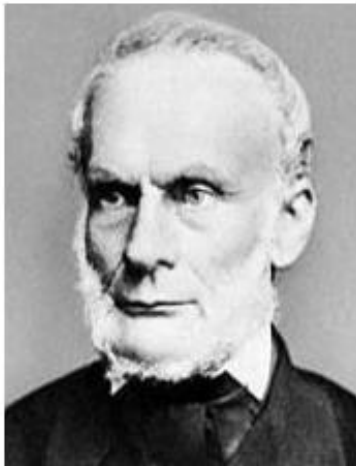
Sadi Carnot



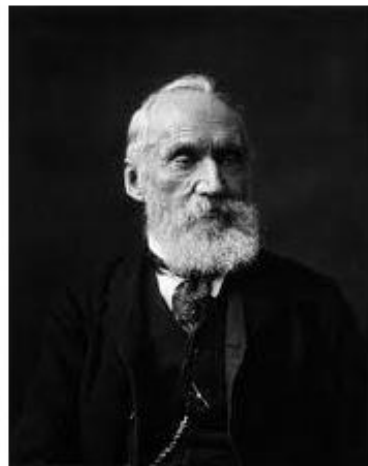
James Prescott Joule



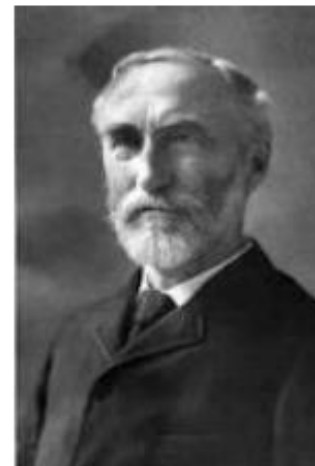
Hermann von Helmholtz



Rudolf J. E. Clausius



William Thomson Kelvin



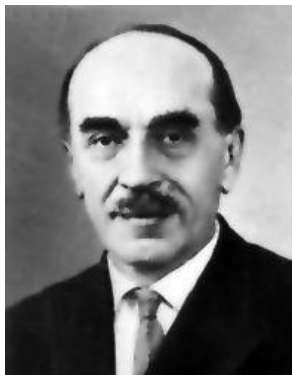
J Willard Gibbs

Introduction of Chemical Kinetics



Modern reaction kinetics based on elementary reactions and radicals, founded by

1920s



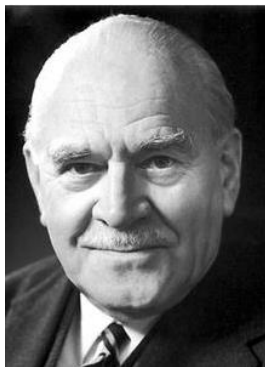
Semenov



Hinshelwood

- Chain reaction and thermal ignition
- The Nobel Prize in Chemistry 1956

1940s



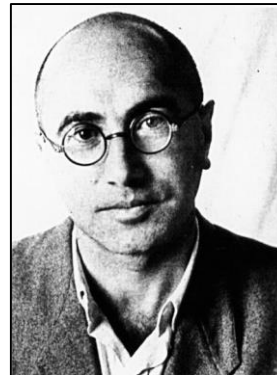
Norrish



Porter

- Extremely fast chemical reactions and hydrocarbon combustion theory
- The Nobel Prize in Chemistry 1967

1940-50s



Zeldovich



Frank-Kamenetskii

- Thermal ignition and pollutant formation

Flame Dynamics and Combustion Physics



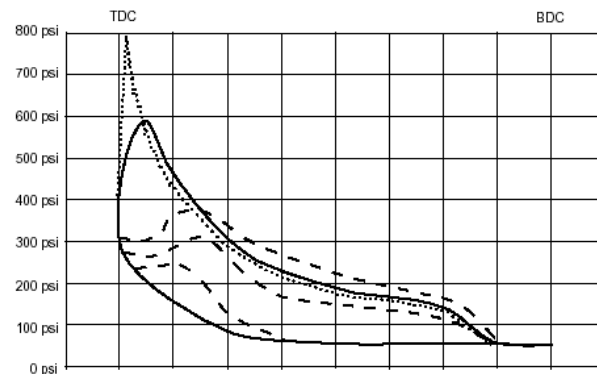
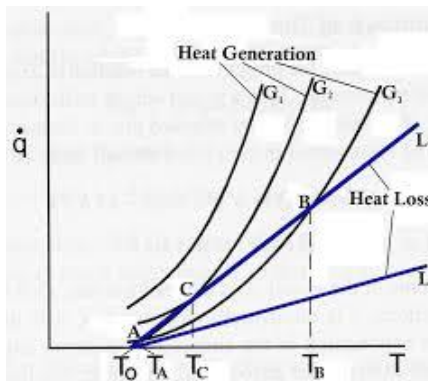
- In 1930s, Lewis et al. began to build the theory of flame dynamics, which put forward some basic concepts of flame physics, such as the minimum ignition energy, flame propagation.
- 1930s-1950s, the law of ignition, flame extinction, flame propagation and turbulent combustion theory was built.



Bernard Lewis
(1899-1993)



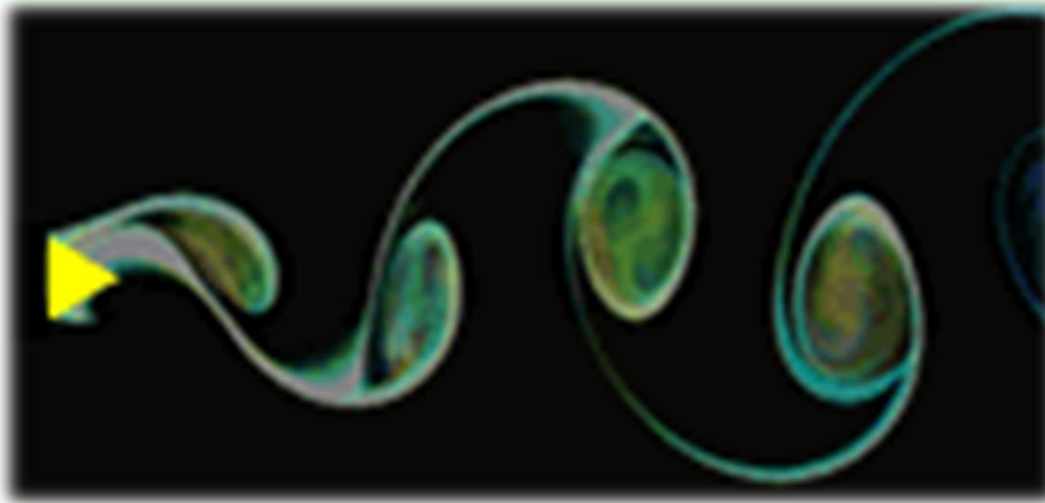
D. A. Frank-Kamenetskii
(1910-1970)



Combustion Theory



von Karman (冯卡门) introduced fluid mechanics to study fundamental combustion process.



Karman vortex streets



Theodore von Kármán (1881-1963)

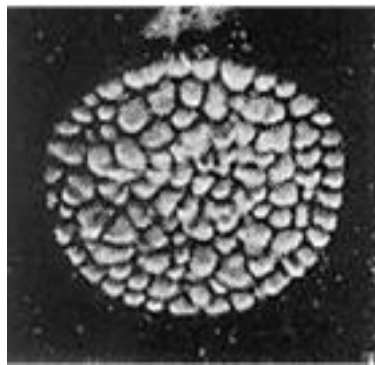
Combustion Theory



Combustion theory developed rapidly from 1960s to 1980s



Ya. B. Zel'dovich



G. H. Markstein



G. I. Sivashinsky



A. Liñán



P. Clavin



F. A. Williams



J. D. Buckmaster

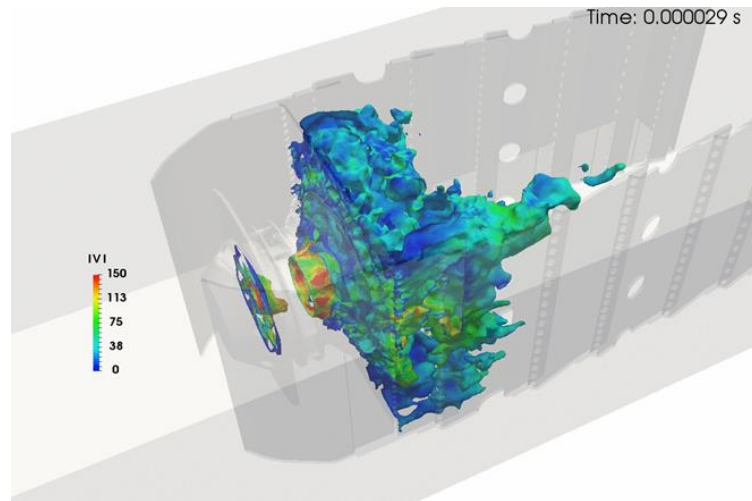
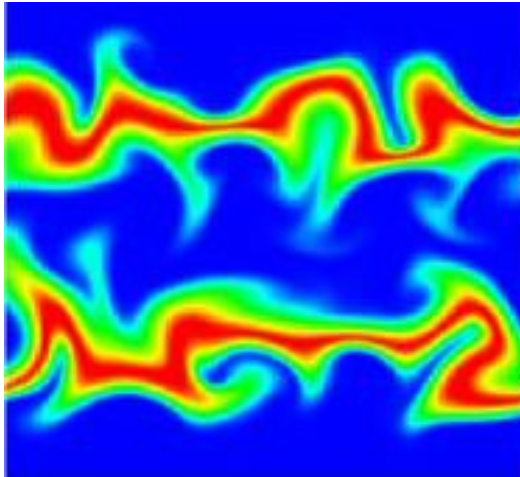


R. W. Bilger



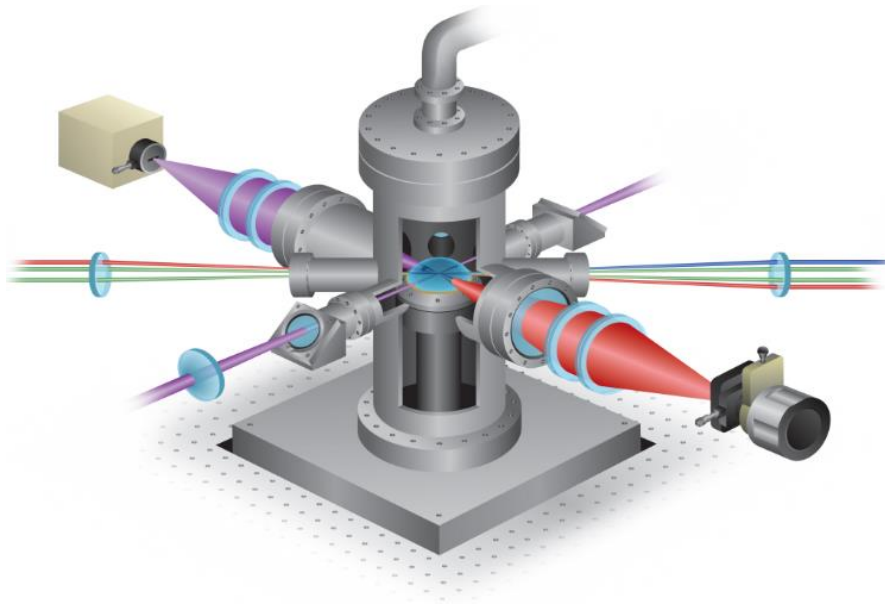
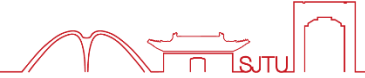
N. Peters

Computational Combustion



- Spalding and colleagues at Imperial College did the pioneering work of building the first combustion simulations on the CFD
- Computational combustion develops rapidly since 1980s

Experimental Combustion



B. Lewis



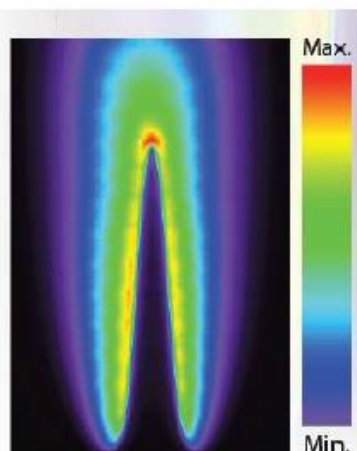
S. Candel



M. Alden

- From 1960s, laser techniques were introduced for combustion research. Laser techniques enable detecting and quantifying important gas-phase species, temperature, and particulate matter in temporally and spatially resolved measurements.

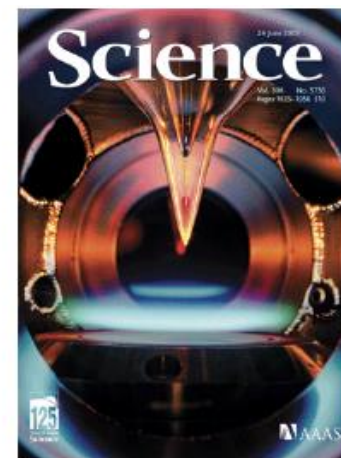
Modern Combustion Research



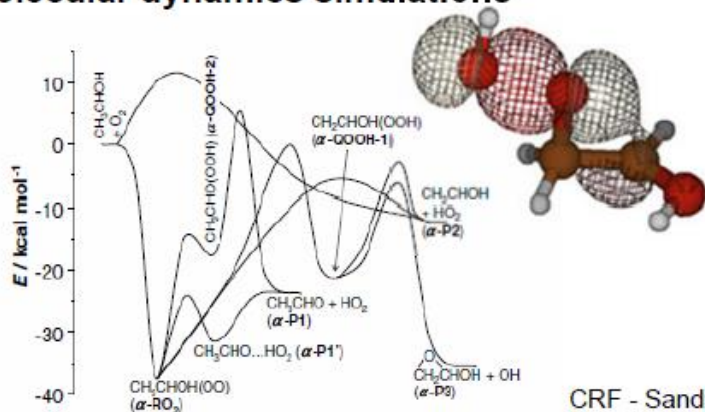
Interference-free composite H-atom LIF image produced with ps excitation in a $\Phi=1.2$ premixed CH₄/O₂/N₂ flame.

CRF - Sandia

Molecular-beam flame sampling and synchrotron-photoionization mass spectrometry analysis
[Science - 24 June 2005]

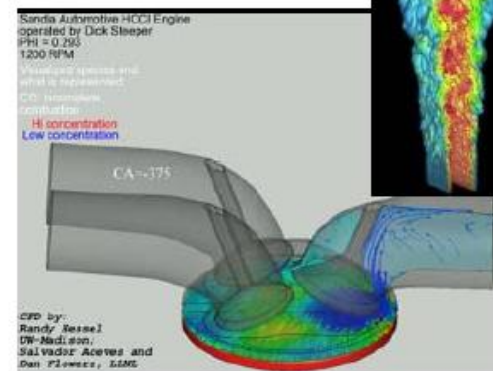


Ab initio electronic-structure calculations and molecular dynamics simulations

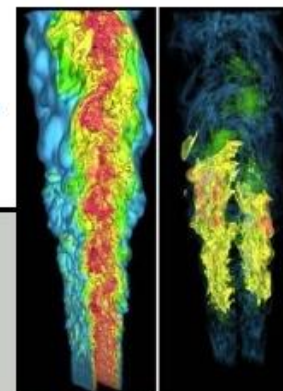


CRF - Sandia

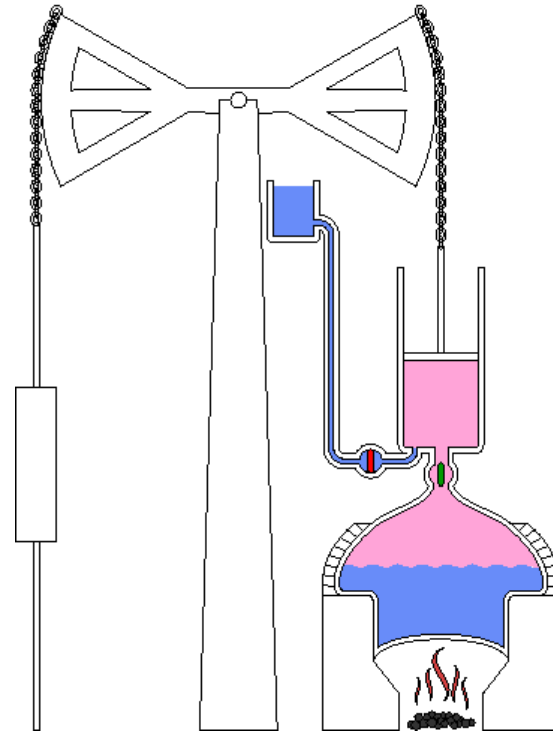
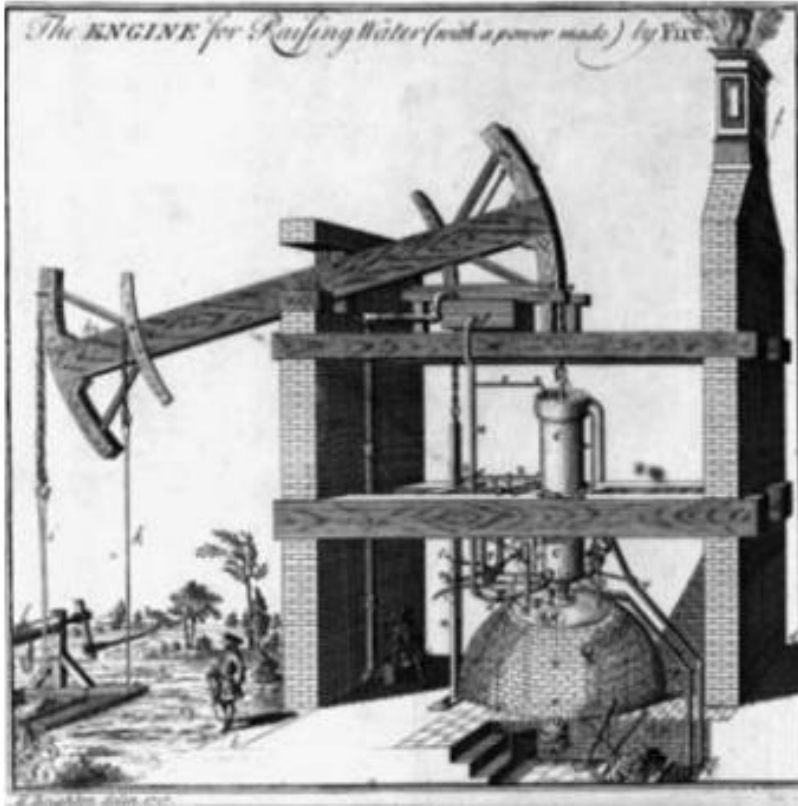
CFD calculations: RANS, LES, DNS. Coupling of numerical chemistry and fluid dynamics



CRF - Sandia



Steam Engine



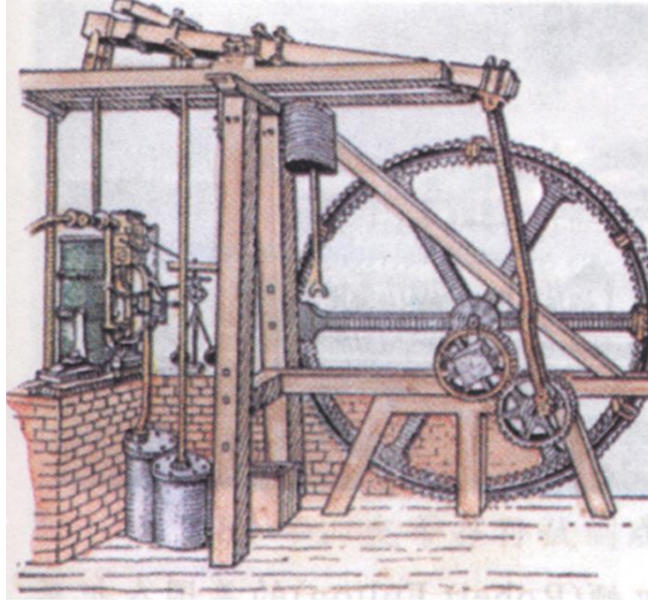
- Thomas Newcomen's atmospheric engine was the first commercial true steam engine using a piston, and was used in 1712 for pumping in a mine.

Steam Engine

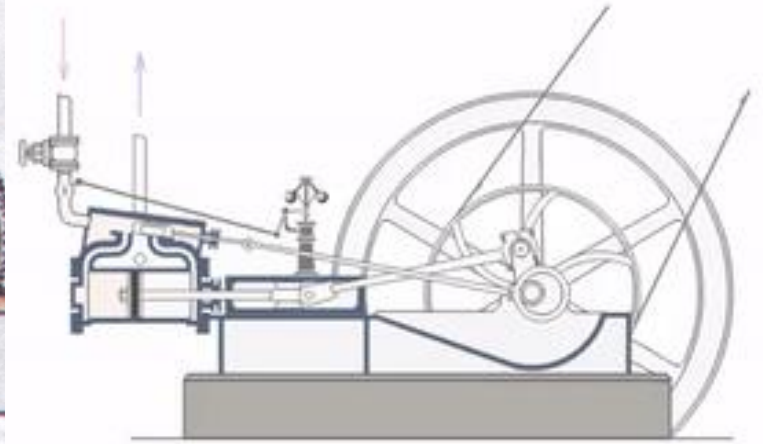
James Watt



Portrait of James Watt (1736–1819)

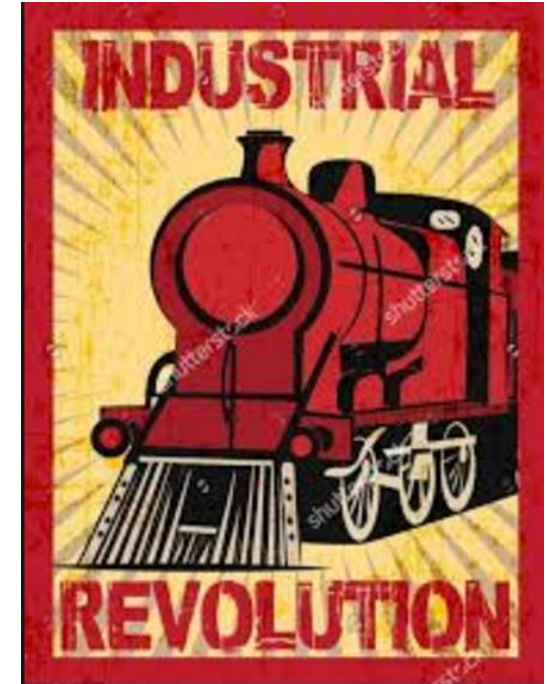
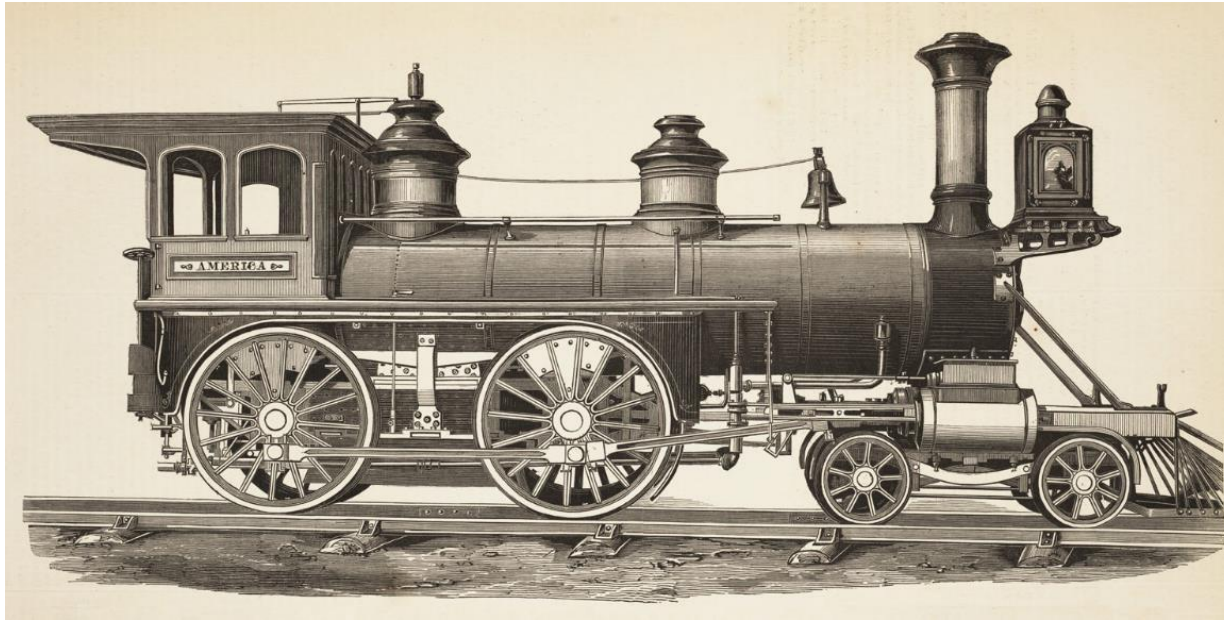


瓦特改良的蒸汽机



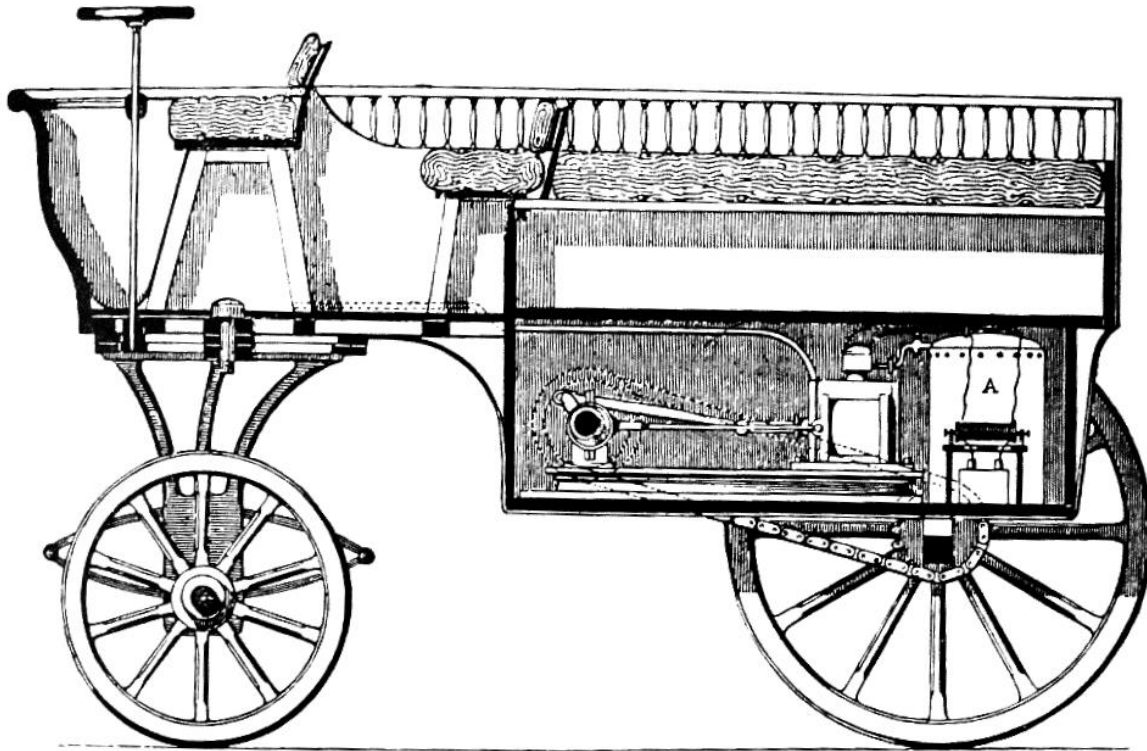
- In 1781 James Watt patented a steam engine that produced continuous rotary motion.
- Watt's ten-horsepower engines enabled a wide range of manufacturing machinery to be powered. The engines could be sited anywhere that water and coal or wood fuel could be obtained.

Steam Engine



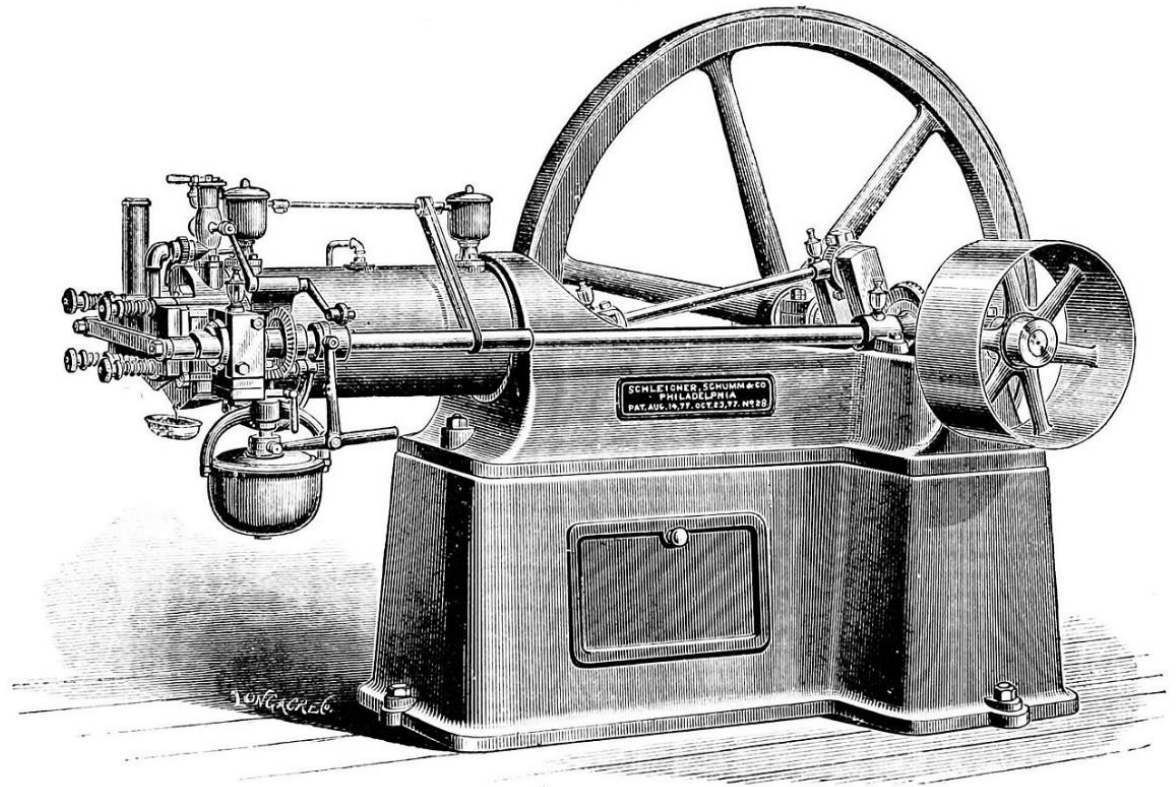
- The stationary steam engine was a key component of the Industrial Revolution, allowing factories to locate where water power was unavailable.

Internal Combustion Engine



- In 1859, Lenoir developed the first internal combustion engine which burned a mixture of coal gas and air ignited by a "jumping sparks" ignition system by Ruhmkorff coil.

Internal Combustion Engine

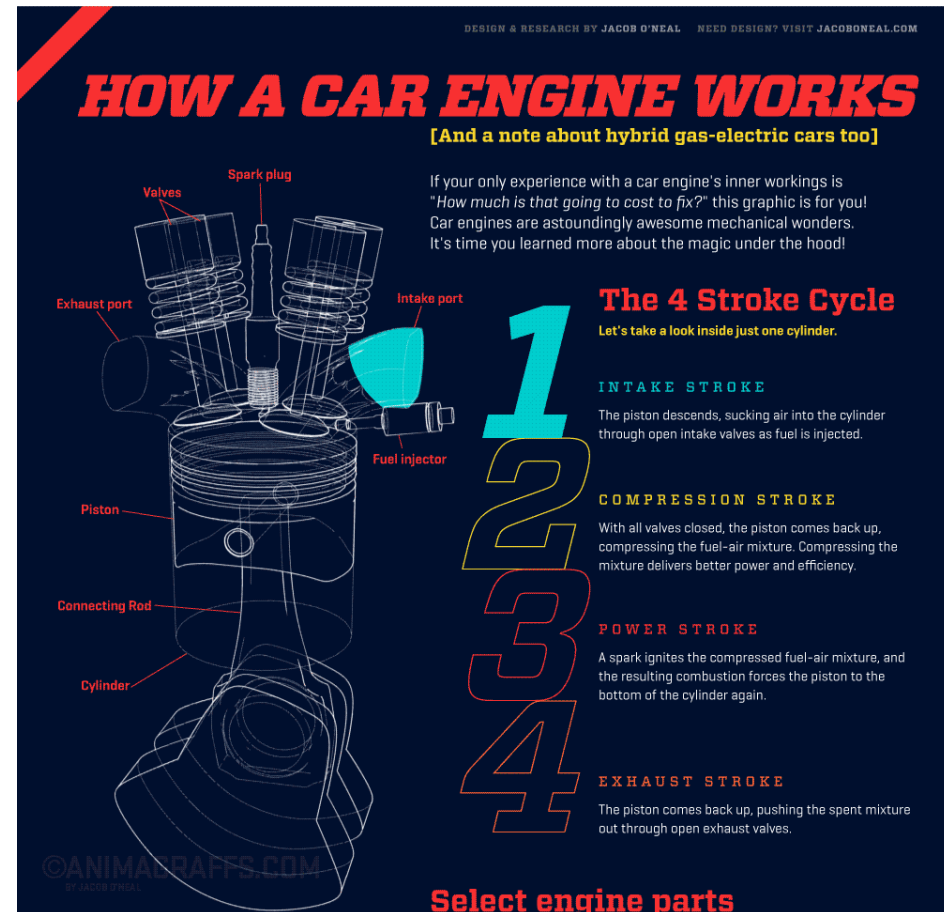


- In 1876, Nikolaus Otto successfully developed the compressed charge internal combustion engine which ran on petroleum gas and led to the modern internal combustion engine.

Internal Combustion Engine



- Typically an ICE is fed with fossil fuels like natural gas or petroleum products such as gasoline, diesel fuel or fuel oil.
- There is a growing usage of renewable fuels like biodiesel for compression ignition engines and bioethanol or methanol for spark ignition engines.



Outline of Chapter 1



- **Combustion with human being**
- **Motivation to study combustion**
- **The history of combustion science and technology**
- **Definition of combustion**
- **Combustion research methods**
- **Introduction of CI and publications**

Definition of Combustion



Webster's Dictionary provides a useful starting point for a definition of **COMBUSTION** as “*rapid oxidation generating heat, or both light and heat; also, slow oxidation accompanied by relatively little heat and no light.*” For our purposes, we will restrict the definition to include only the rapid oxidation portion, since most practical combustion devices belong in this realm.

Principle of Combustion



H. W. Emmons
(1912-1998)

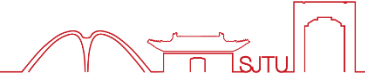
Combustion phenomena are basically
chemical reactions with rapid heat production.

H. W. Emmons, Proc. Combust. Inst. 13 (1971) 1-18.

- ❑ Combustion processes: convert chemical energy to thermal energy
- ❑ A complicated sequence of **elementary radical reactions**



Principle of Combustion



Combustion: chemically reactive flow

➤ Thousands of species
and tens of thousands



been used for
diagnostics.

- ignition
- extinction
- pollutants
- propagation
-



- ◆ reactions
- ◆ turbulent flow
- ◆ mass transport
- ◆ heat radiation

1 Pyrolysis

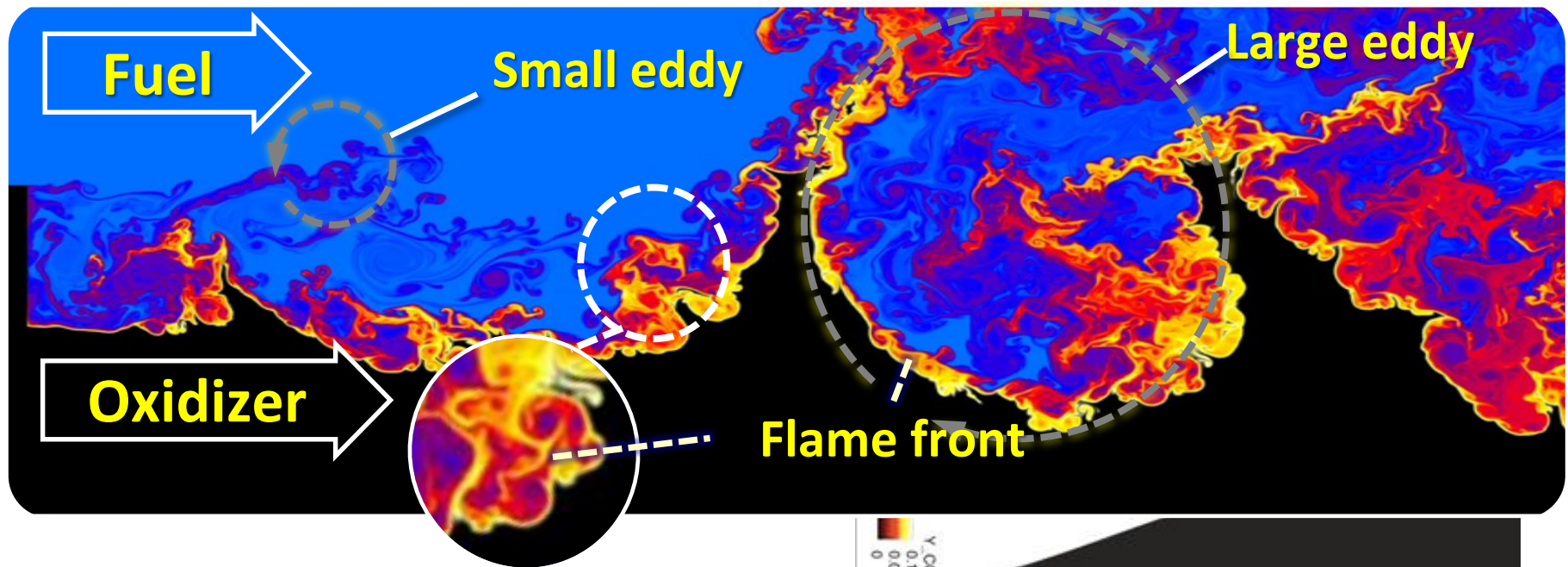
2 Low-T oxidation

3 High-T oxidation

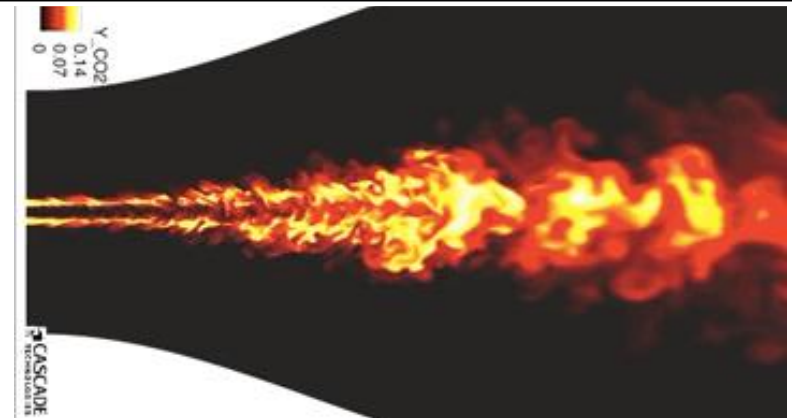
Principle of Combustion



Complex flow patterns



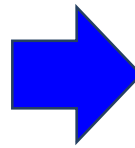
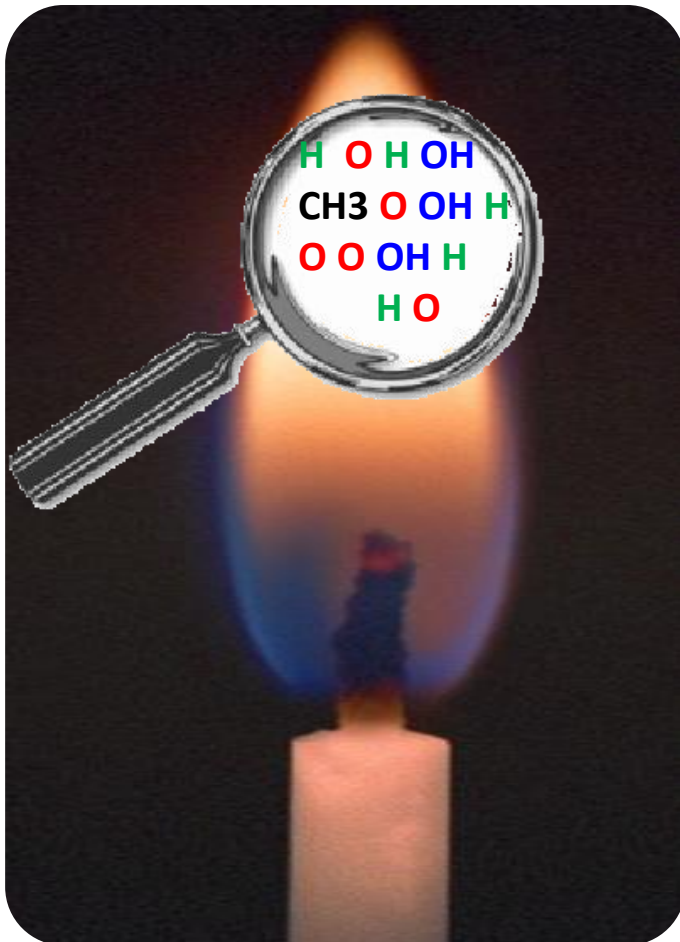
- Complex turbulent flow
- Complex physico chemistry process



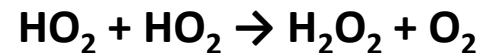
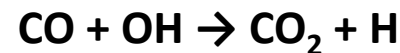
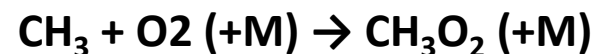
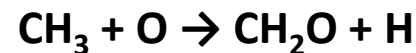
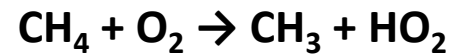
Principle of Combustion



Complex chain reactions



Chain Reactions



Flame Structure



3D flame



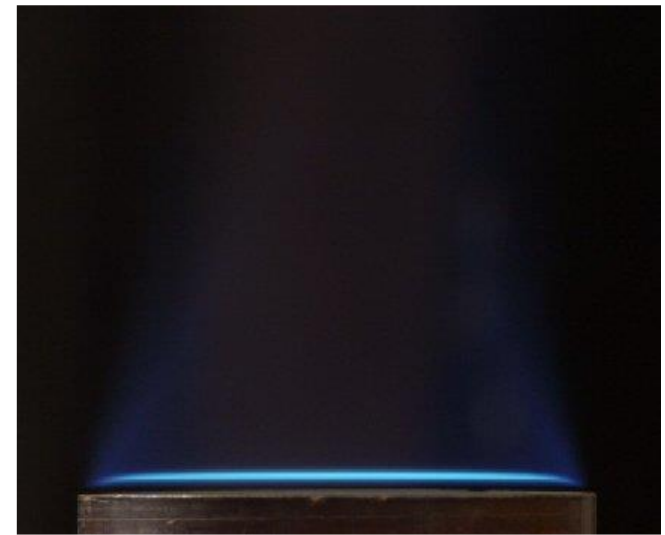
Temperature and species concentration are the functions of three spatial coordinates

2D flame



Temperature and species concentration are the functions of two spatial coordinates
(**axisymmetric flame**)

1D flame



Temperature and species concentration are the functions of one spatial coordinates

Flame Types



	premixed	non-premixed
laminar flame	gas stove flame Bunsen burner with blue flame	candle Bunsen burner with yellow flame
turbulent flame	petrol engine with carburator	gas turbine rocket engine ($\text{H}_2 + \text{O}_2$ or kerosene + O_2) Diesel engine

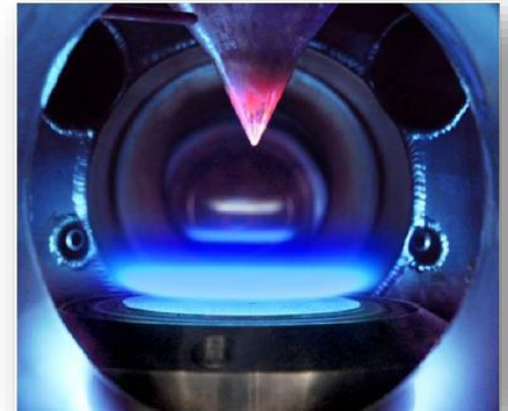
Flame Types



Premixed and non-premixed are related to the state of mixedness of the reactants

- Premixed flame

- ✓ Mixing and then burning;
- ✓ It is safe only in small scale; not used in the industry.

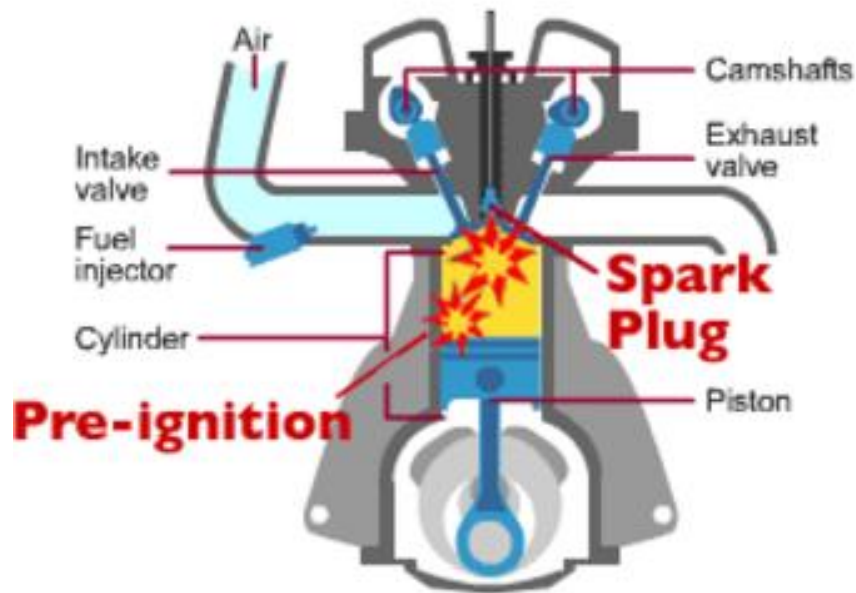


- Non-premixed flame

- ✓ Mixing and burning simultaneously;
- ✓ Safe, because the combustible mixture is generated locally.



Flame Types



- ❑ The spark-ignition engine is an example where premixed flames occur.
- ❑ Diesel-engine combustion is generally considered to have significant amounts of both premixed and non-premixed flames.

Flame Types



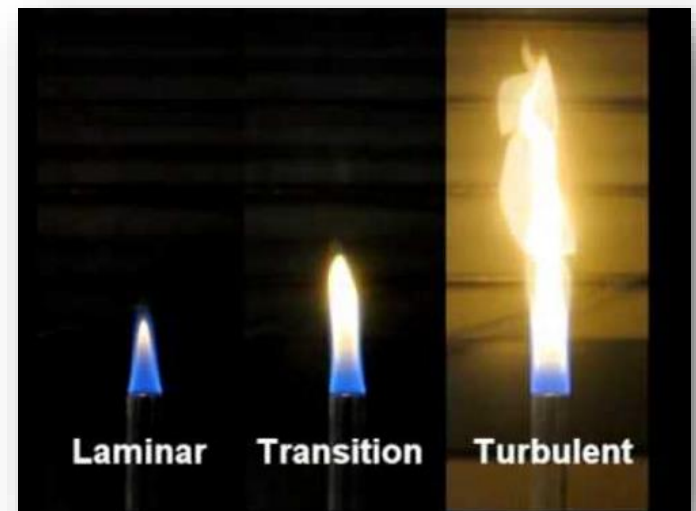
- **Laminar flame**

- ✓ The total heat production is small but large heat production in some points.



- **Turbulent flame**

- ✓ Large fuel consumption, large total heat production but small heat production in each point!





The Combustion Processes



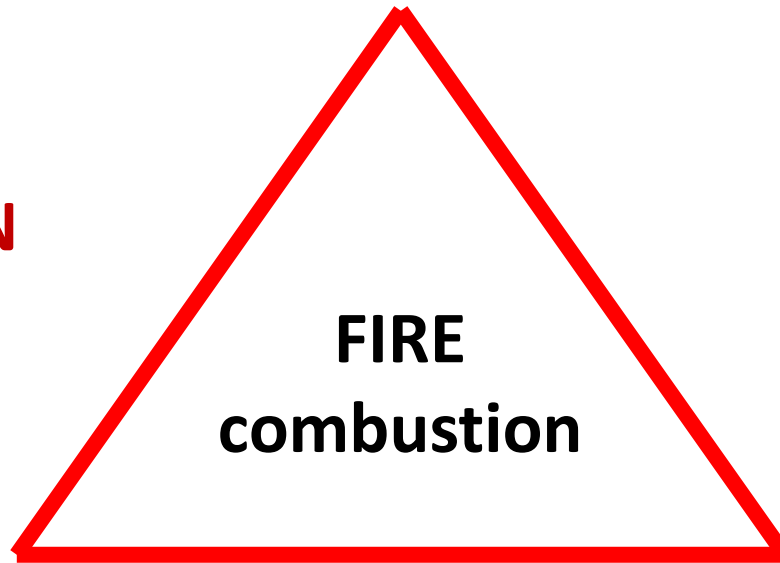
THE FIRE TRIANGLE

OXYGEN

HEAT

**FIRE
combustion**

FUEL

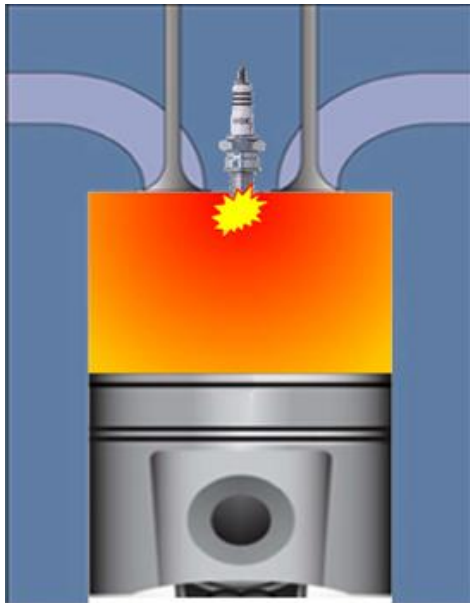


The Combustion Processes

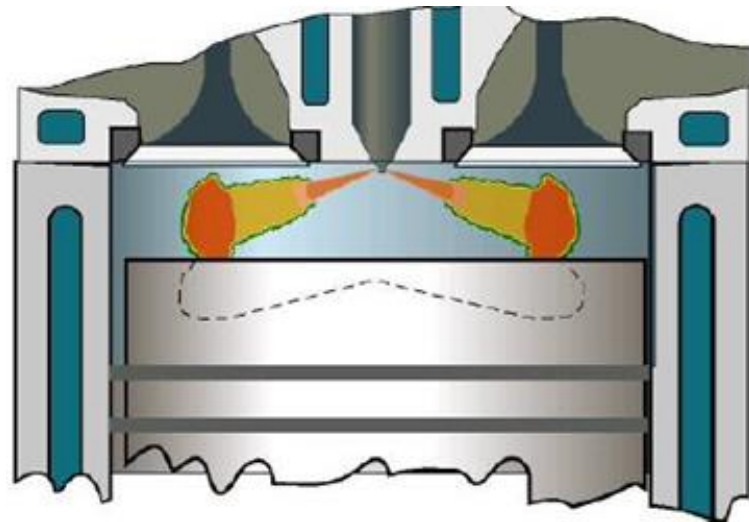


Ignition

piloted ignition



spontaneous ignition

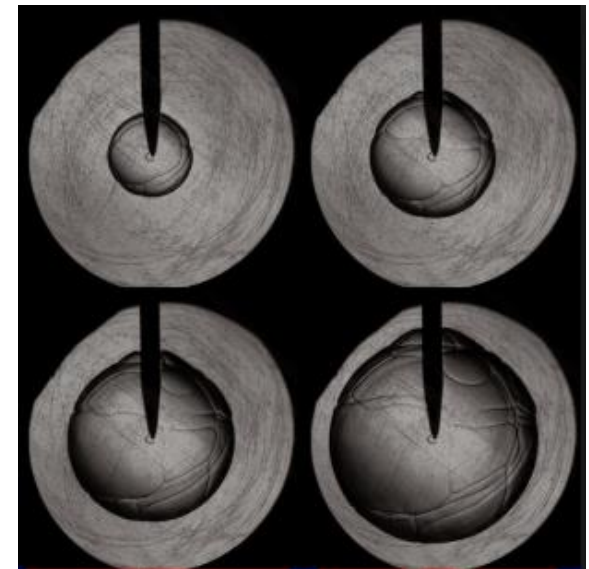
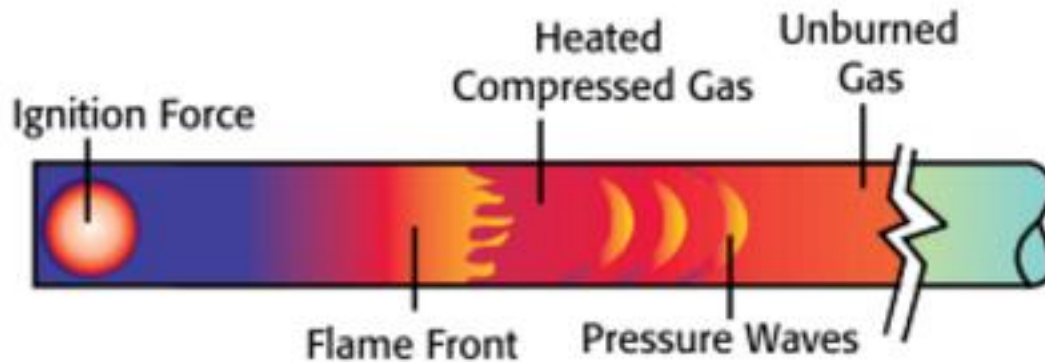


- ❑ Ignition is the mechanism leading to the onset of a vigorous combustion reaction and is characterized by a rapid increase of the species temperature.

The Combustion Processes



Combustion and flame propagation



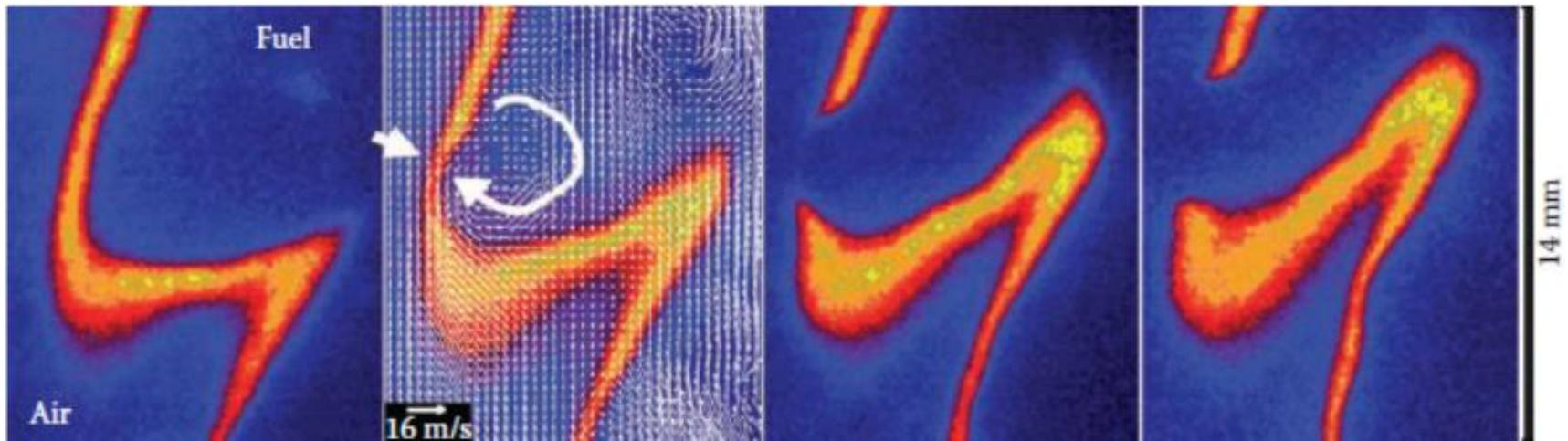
- ❑ Flame propagation from unburnt gas to burnt gas.
- ❑ laminar flame velocity is a physical constant, at a given gas composition, initial temperature and pressure.

The Combustion Processes



Extinction

localized extinction event in a turbulent non-premixed jet flame

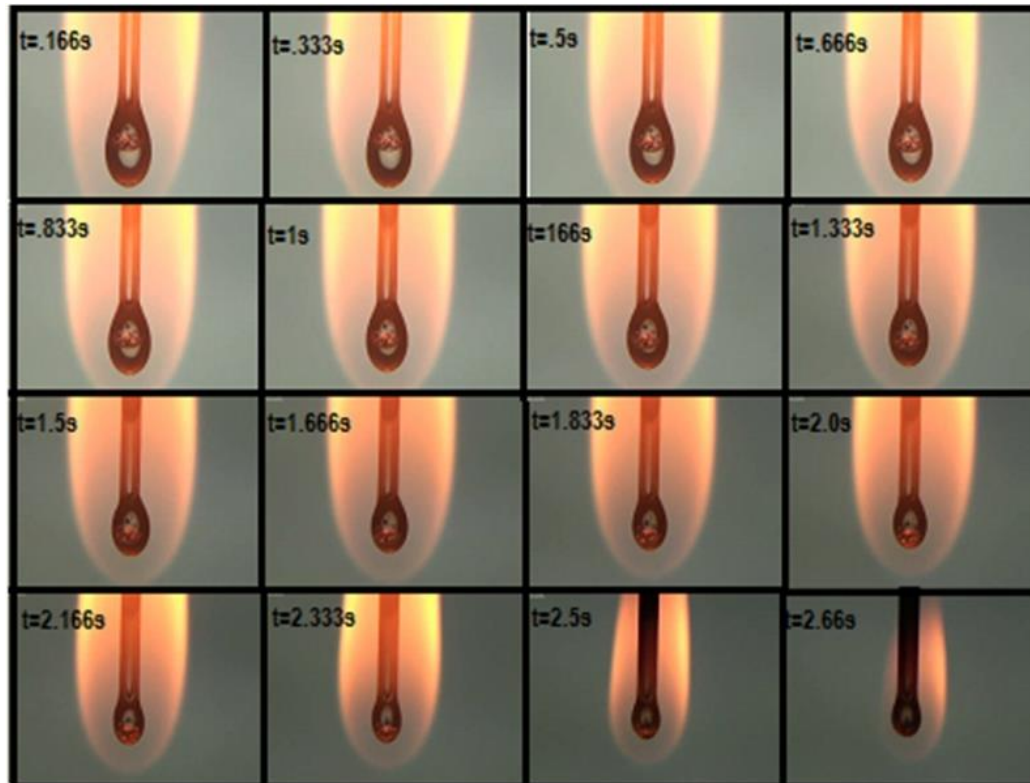


- ❑ Flame extinction is related to heat loss in flames and the changes of flame conditions.

The Combustion Processes



Droplet Combustion



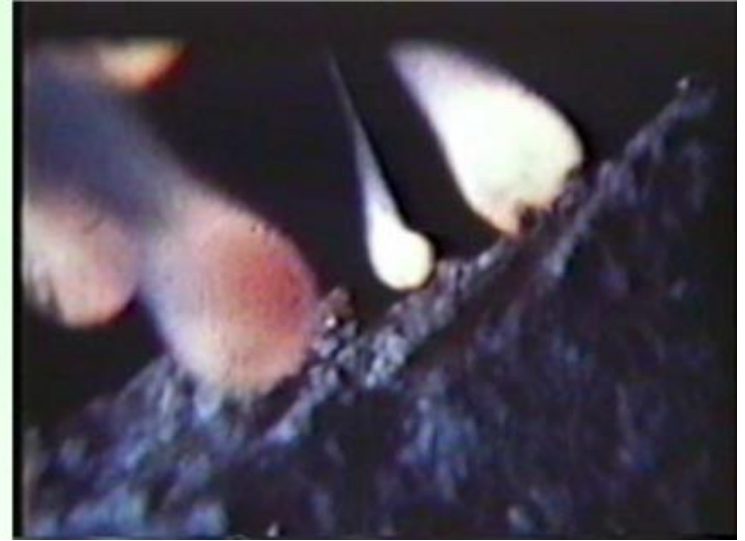
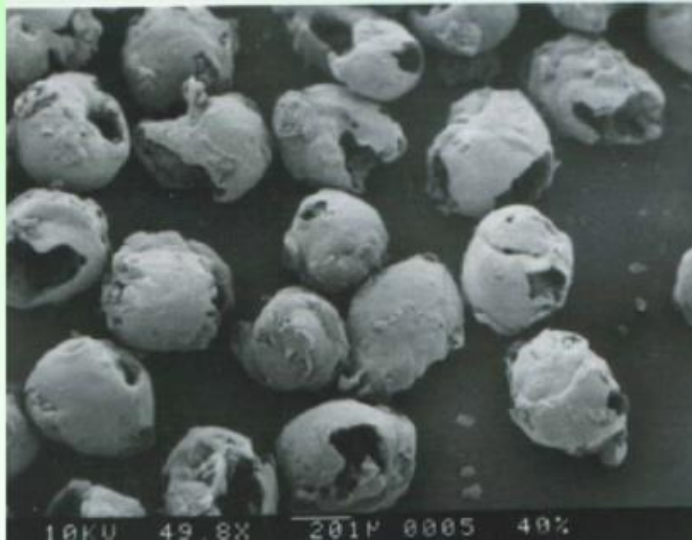
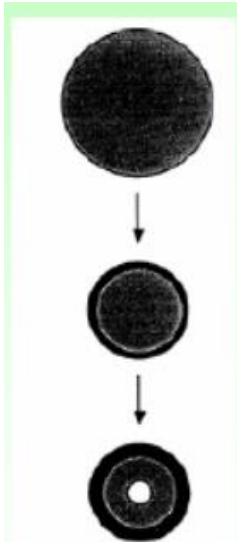
- ❑ Droplet combustion is important in internal combustion engines and jet engines.

The Combustion Processes



Solid Combustion

Coal, biomass, metal, metal oxides...

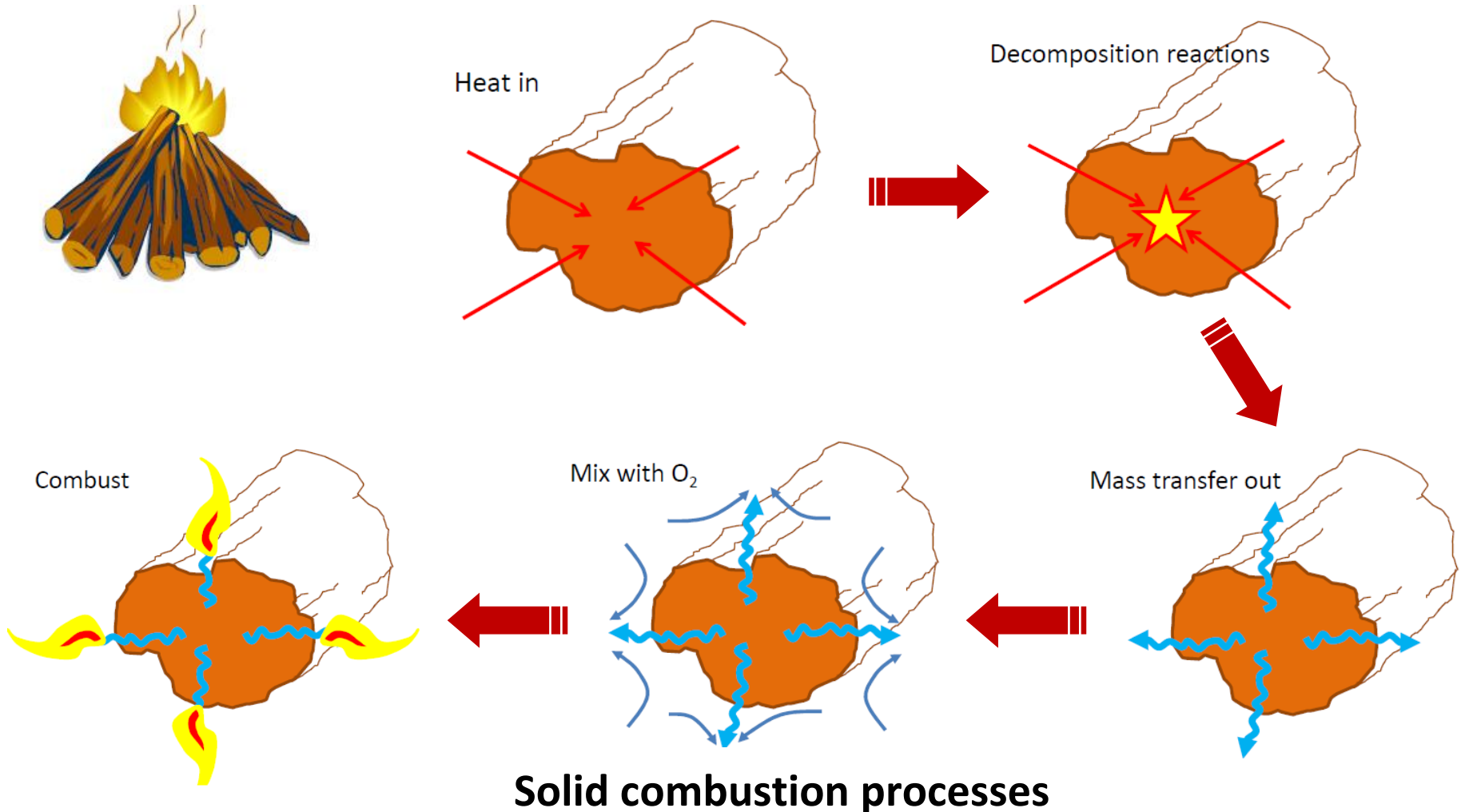


❑ Combustion of solid fuels is a complex physico-chemical process.

The Combustion Processes



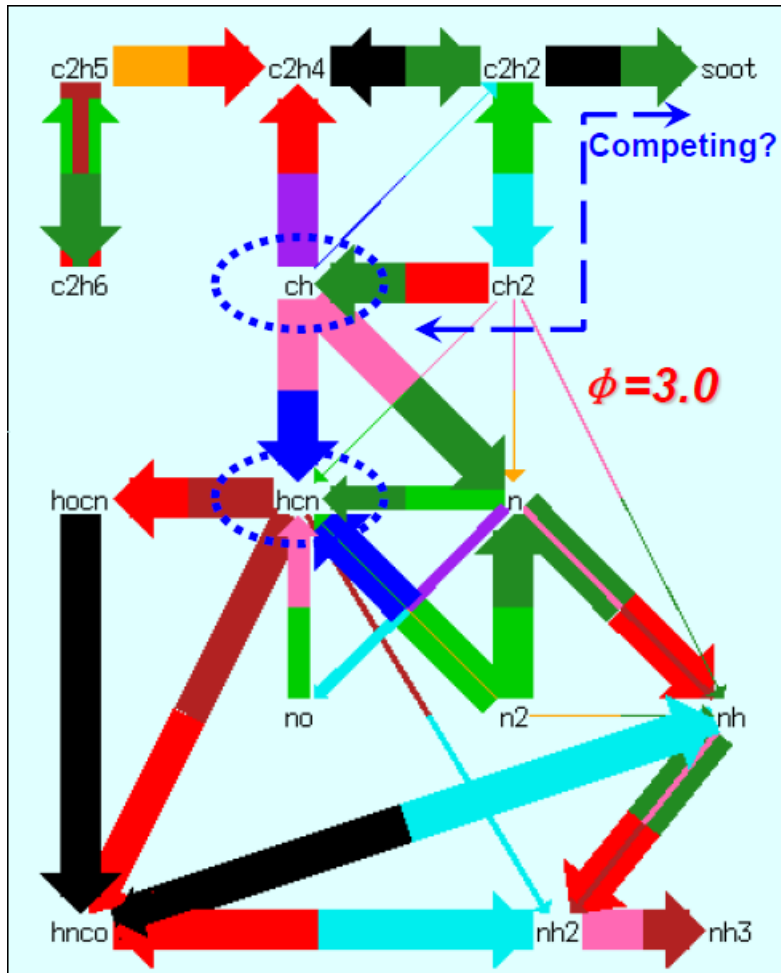
Solid Combustion



The Pollutant Formation Processes

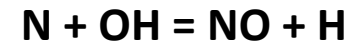
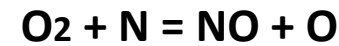
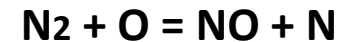


NOx formation

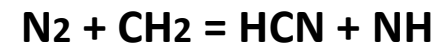
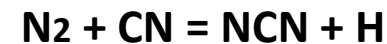


Formation mechanism

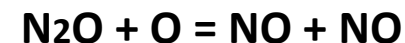
◆ Thermal NO



◆ Prompt NO



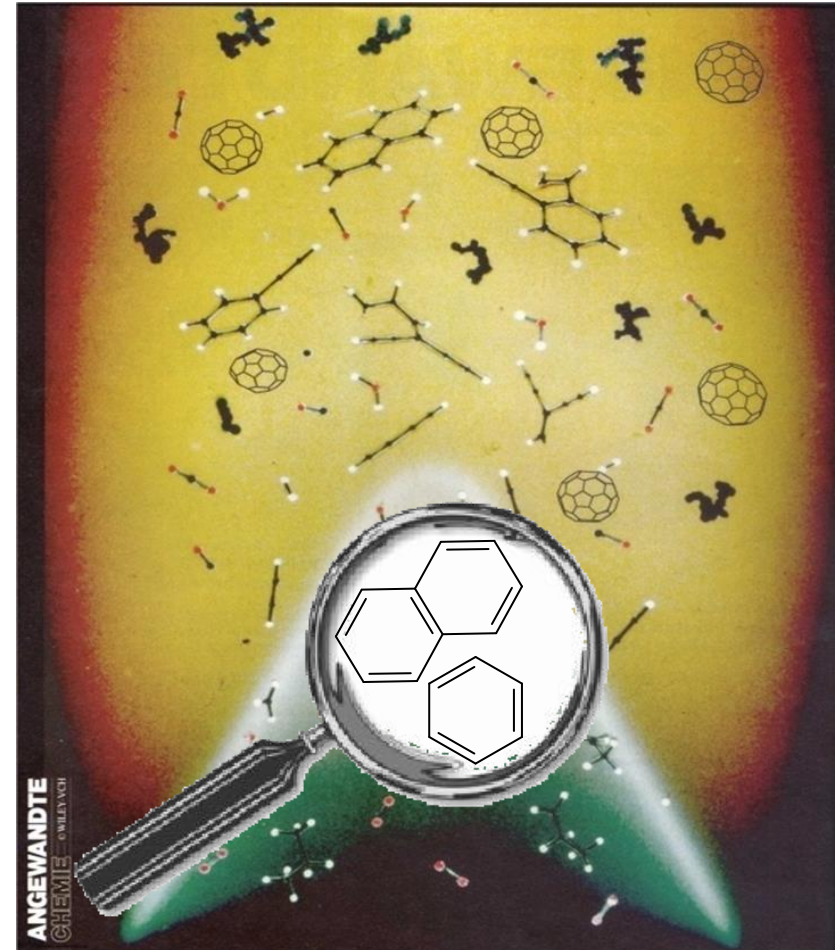
◆ N₂O mechanism



The Pollutant Formation Processes



PAH and soot formation

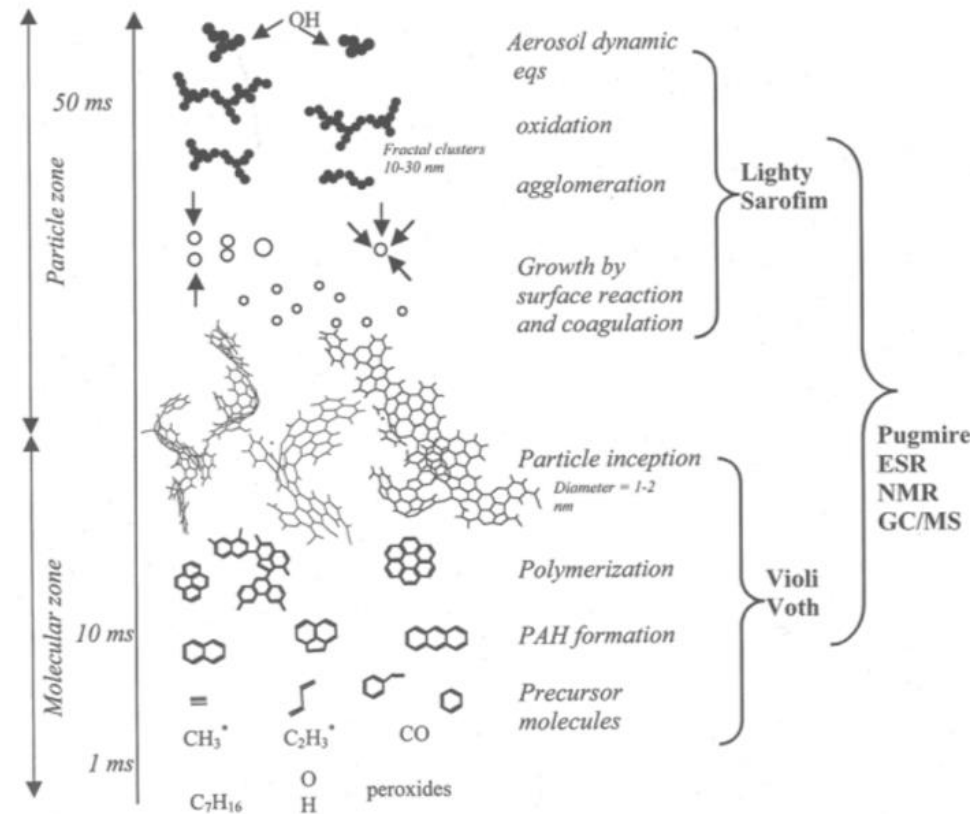


The Pollutant Formation Processes



PAH and soot formation

- The formation of soot is a complex process, an evolution of matter in which a number of molecules undergo many chemical and physical reactions within a few milliseconds.



Outline of Chapter 1



- **Combustion with human being**
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Characteristic of Combustion Science



Crossing of many disciplines

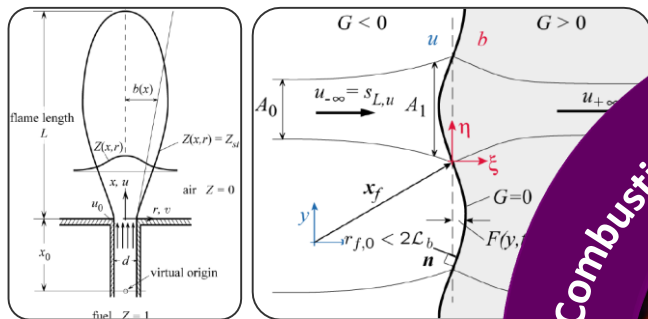


- ☐ Thermodynamics
- ☐ Chemical Kinetics
- ☐ Fluid Mechanics
- ☐ Heat Transfer
- ☐ Mass Transfer
- ☐ Physical Chemistry
- ☐

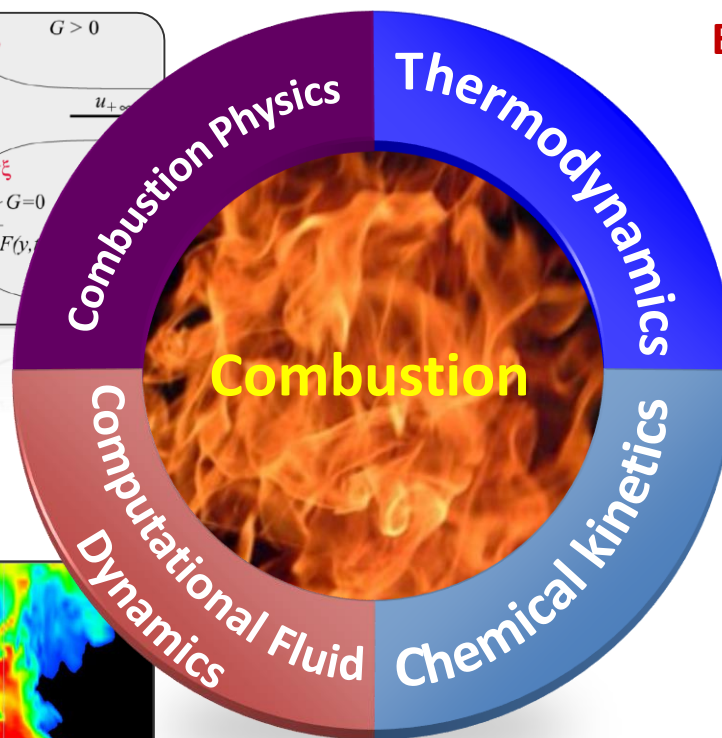
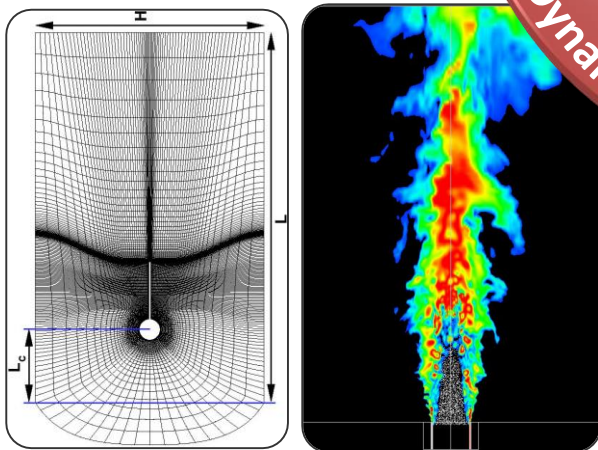
Research Method



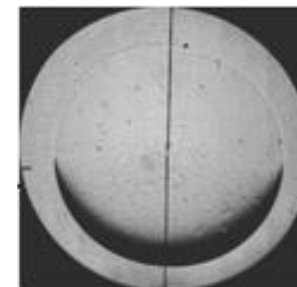
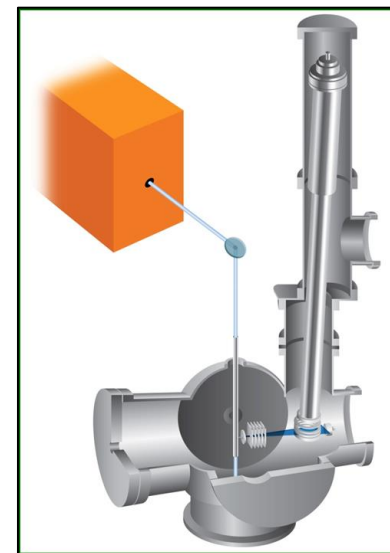
Theoretical analysis



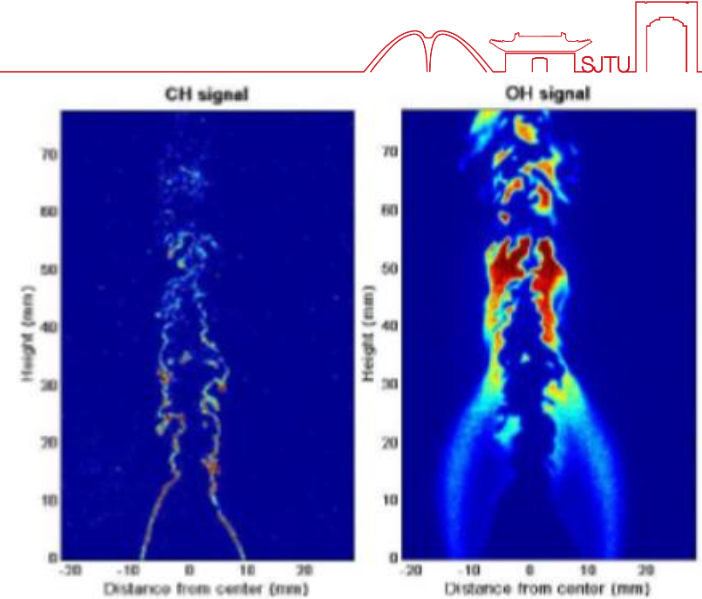
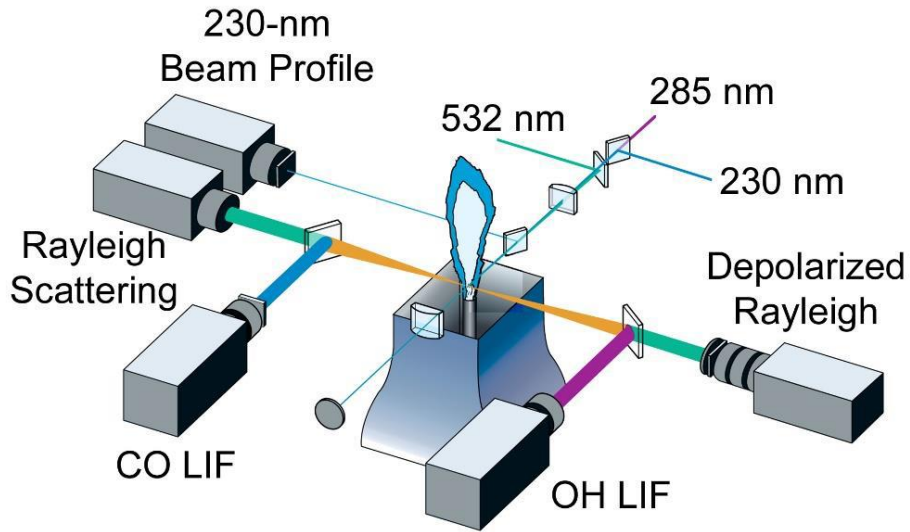
Numerical simulation



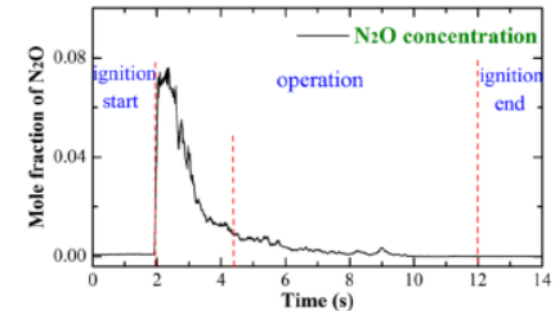
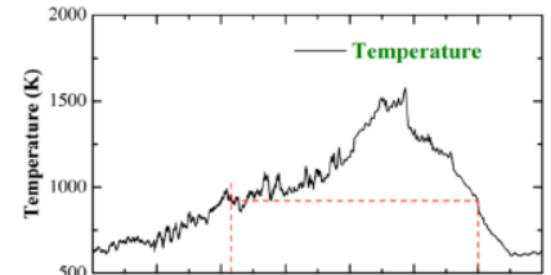
Experimental measurement



Experimental Measurement



- Temperature, velocity, pressure
- Species concentration
- Global combustion parameters
-



Theoretical Analysis



$$[\rho' \bar{u} + u' \bar{\rho}]_1^2 = 0,$$

$$[p' + \rho' \bar{u}^2 + 2\bar{\rho} \bar{u} u']_1^2 = 0,$$

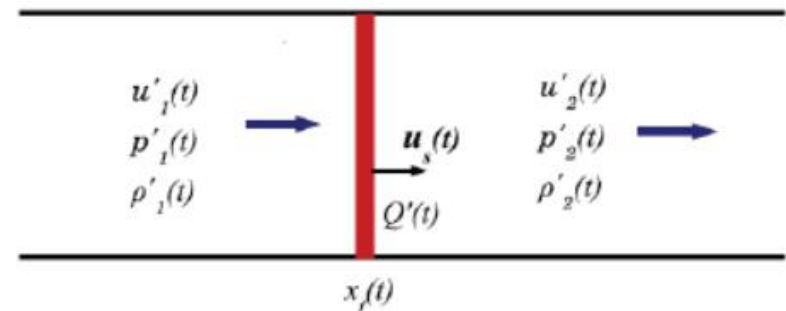
$$[c_p \bar{T} (\rho' \bar{u} + u' \bar{\rho}) + \bar{\rho} \bar{u} (c_p T' + \bar{u} u')]_1^2 = \dot{Q}'.$$

$$\rho_2 u_2 - \rho_1 u_1 = u_s (\rho_2 - \rho_1)$$

$$\rho_2 u_2^2 - \rho_1 u_1^2 + p_2 - p_1 = u_s (\rho_2 u_2 - \rho_1 u_1)$$

$$\rho_2 u_2 H_2 - \rho_1 u_1 H_1 - \dot{Q} = u_s (\rho_2 E_2 - \rho_1 E_1)$$

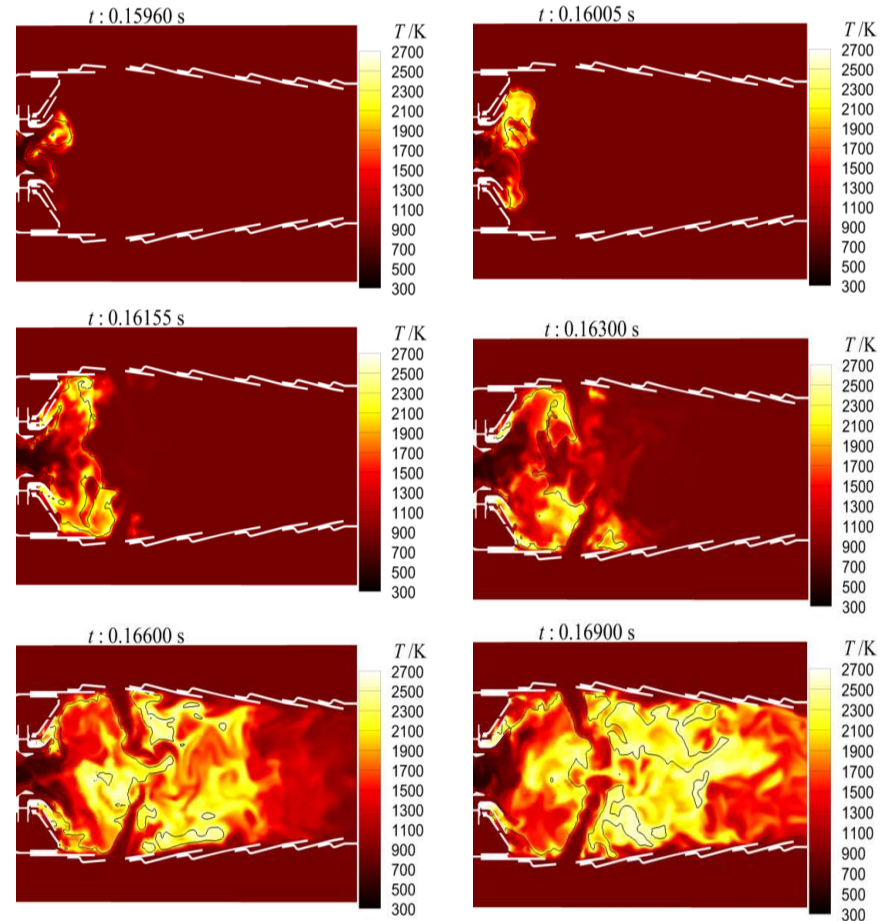
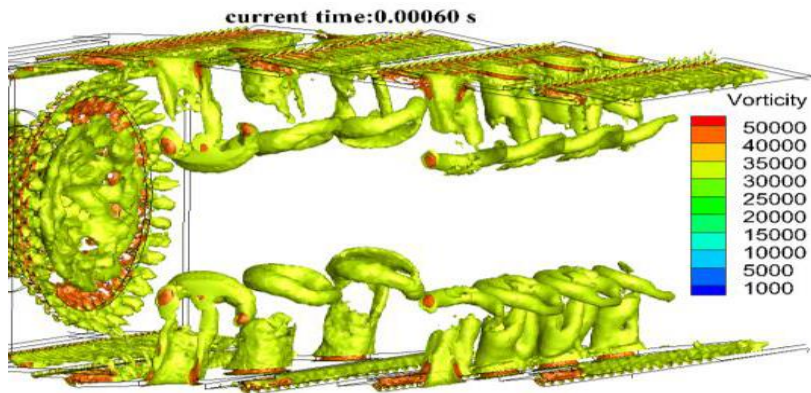
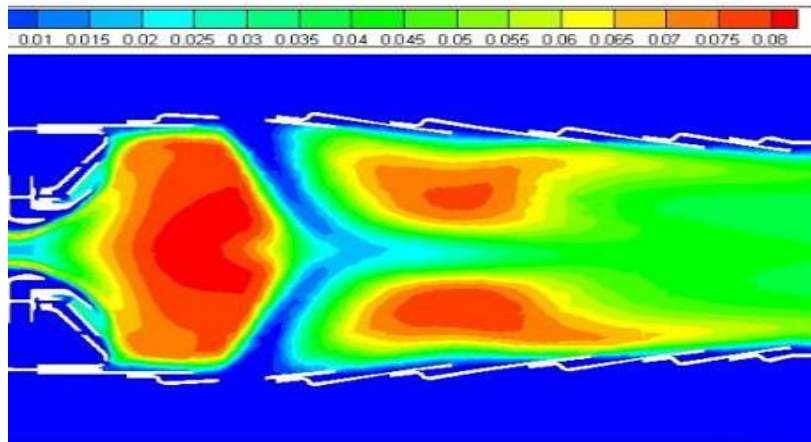
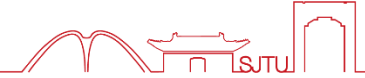
$$\frac{s'_2}{c_p} = \underbrace{\left(1 - \frac{1}{\lambda}\right) \frac{\phi'}{\bar{\phi}}}_{\mathcal{O}(1)} + \underbrace{\left(1 - \frac{1}{\gamma}\right) \left(\frac{1}{\lambda} - 1\right) \frac{p'_1}{\bar{p}_1}}_{\mathcal{O}(M)} + \underbrace{\frac{s'_1}{c_p}}_0.$$



□ Develop and improve combustion theory

□ Explain experimental observations

Numerical Simulation



- Bridge between combustion theory and experiment
- Predict combustion process in practical combustion devices

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Modern Combustion Research



The international organization of combustion science



❑ The Combustion Institute (CI), Pittsburgh

❑ <https://www.combustioninstitute.org>

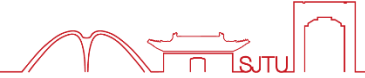
❑ Affiliated Journals:

Combustion and Flame (IF=4.168)

Proceedings of the Combustion Institute (IF=4.120)

Progress in Energy and Combustion Science (IF=16.784)

International Symposium on Combustion



- Sep. 10-14, 1928. The First Symposium on Combustion held at the 76th Meeting of ACS at Massachusetts .
- Sep. 9-10, 1937, The Second Symposium on Combustion held at the 94th Meeting of ACS at New York .
- Sep. 7-11, 1948, The Third Symposium on Combustion at Wisconsin.
Standing Committee on Combustion Symposia
- Sep. 1-5, 1952, The Fourth Symposium on Combustion at Massachusetts.
- Aug. 30-Sep. 3, 1954, The Fifth Symposium on Combustion at Pennsylvania.
- Aug. 19-24, 1956, The Sixth Symposium on Combustion at Connecticut. (the first one held by The Combustion Institute)
- Aug. 28-Sep. 3, 1958, The Seventh Symposium on Combustion at London and Oxford.
-

International Symposium on Combustion



Since 1949, in every second year

1990	23 rd Symposium	<i>Orleans, France</i>
1992	24 th Symposium	<i>Sydney, Australia</i>
1994	25 th Symposium	<i>Irvine, USA</i>
1996	26 th Symposium	<i>Naples, Italy</i>
1998	27 th Symposium	<i>Boulder, USA</i>
2000	28 th Symposium	<i>Edinburgh, UK</i>
2002	29 th Symposium	<i>Sapporo, Japan</i>
2004	30 th Symposium	<i>Chicago, USA</i>
2006	31 st Symposium	<i>Heidelberg, Germany</i>
2008	32 nd Symposium	<i>Montreal, Canada</i>
2010	33 rd Symposium	<i>Beijing, China</i>
2012	34 th Symposium	<i>Warsaw, Poland</i>
2014	35 th Symposium	<i>San Francisco, USA</i>
2016	36 th Symposium	<i>Seoul, South Korea</i>
2018	37 th Symposium	<i>Dublin, Ireland</i>

Colloquia of Combustion Symposium



1. **GAS-PHASE REACTION KINETICS** including the kinetics of hydrocarbons and oxygenated fuels, formation of gaseous pollutants, elementary reactions, mechanism generation, reduction and uncertainty quantification.
2. **SOOT, NANOMATERIALS, AND LARGE MOLECULES** including the formation, growth, and destruction of soot, PAHs, carbon nanostructures, and other nanoscale materials.
3. **DIAGNOSTICS** including the development and application of diagnostic techniques and sensors for the understanding and control of combustion and reacting flow phenomena.

Colloquia of Combustion Symposium



4. **LAMINAR FLAMES** including their ignition, structure, propagation, extinction, stabilization, dynamics, and instabilities.
5. **TURBULENT FLAMES** including their ignition, structure, propagation, extinction, stabilization, dynamics, and instabilities.
6. **SPRAY, DROPLET, AND SUPERCRITICAL COMBUSTION** including atomization, combustion of droplets, sprays, and supercritical fluids.
7. **DETONATIONS, EXPLOSIONS, AND SUPERSONIC COMBUSTION** including flame acceleration, DDT, and pulse-detonation-, constant volume combustion-, and scramjet-engines.

Colloquia of Combustion Symposium



8. **SOLID FUEL COMBUSTION** including fundamental aspects related to pyrolysis, oxidation, gasification, and ash formation from coal, biomass, and wastes, as well as combustion of propellants and metals.
9. **FIRE RESEARCH** including fundamental aspects of ignition, burning, spread and suppression of fire, as well as applications to building fire and urban/wildland fire safety.
10. **STATIONARY COMBUSTION SYSTEMS AND CONTROL OF GREENHOUSE GAS EMISSIONS** including combustion in stationary power generation, fluidized beds, incineration, utility boilers, industrial applications, NO_x and SO_x reduction, MILD combustion, oxy-fuel combustion, chemical looping, and CO₂ capture.

Colloquia of Combustion Symposium

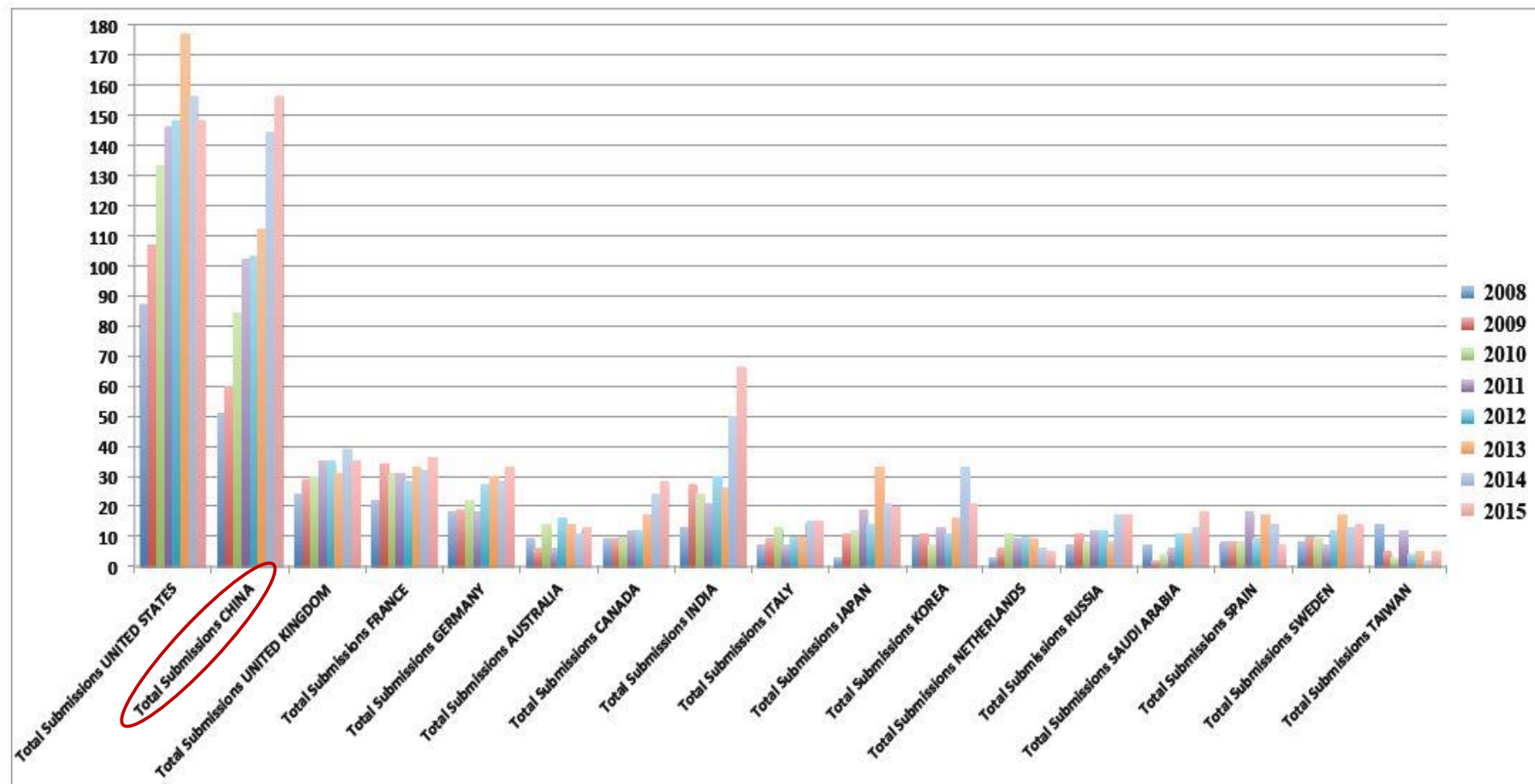


- 11. INTERNAL COMBUSTION ENGINES** including device-specific aspects of fuels research, direct injection, and combustion dynamics (e.g. ignition, quenching).
- 12. GAS TURBINE COMBUSTION** including gas turbines for propulsion and power generation, as well as device-specific aspects of fuels research, burner stability, and combustion dynamics (e.g. ignition, quenching, thermoacoustics).
- 13. OTHER CONCEPTS** including assisted combustion (plasmas, electric and magnetic fields), catalysis, fuel synthesis and transformation, micro-channel reactors, integrated process intensification, fuel cells, and electrolysis.

Publications — CnF



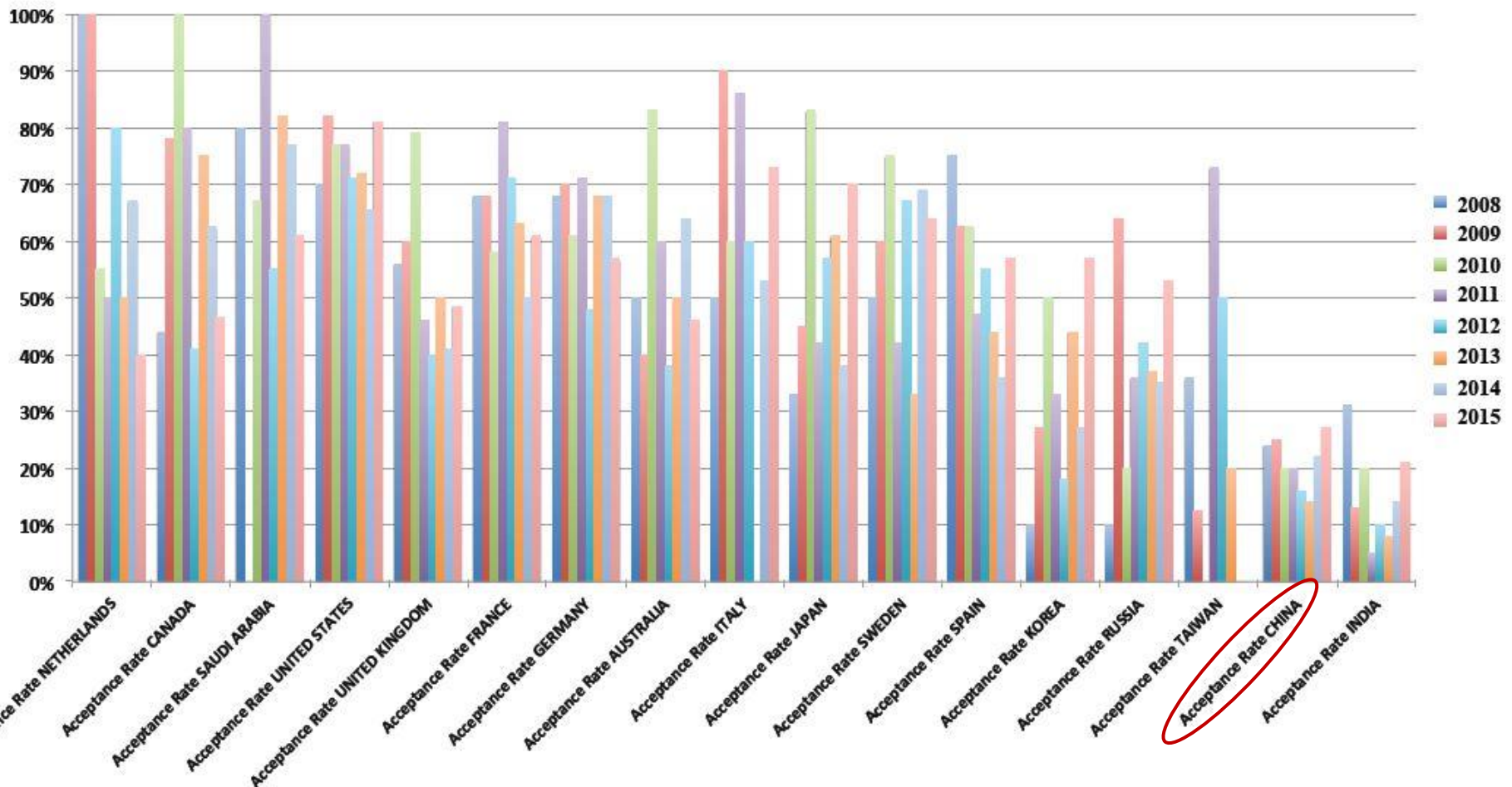
Submissions by Country/Region (2008-2015)



Publications — CnF



Acceptance Rates by Country/Region (2008-2015)



Publications — the 36th ProCI by Country/Region



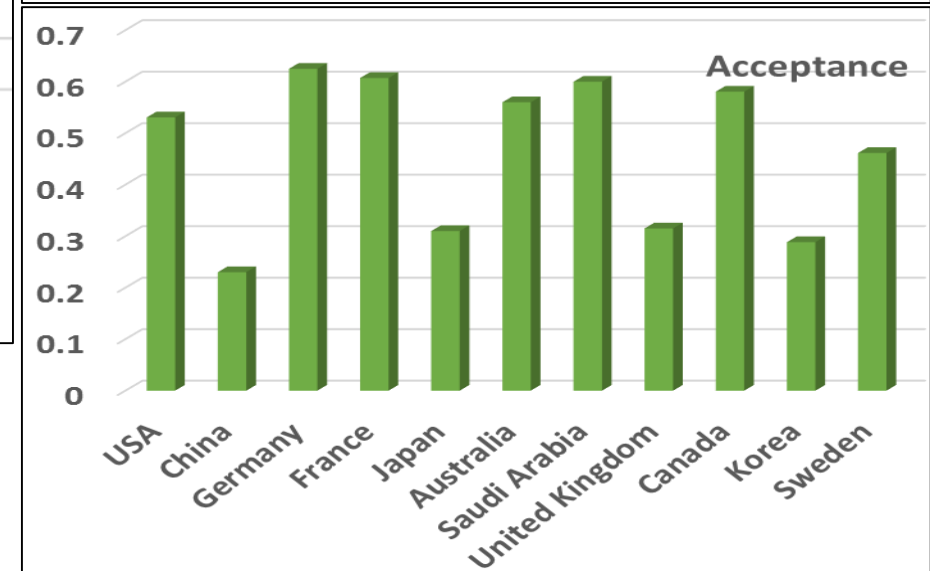
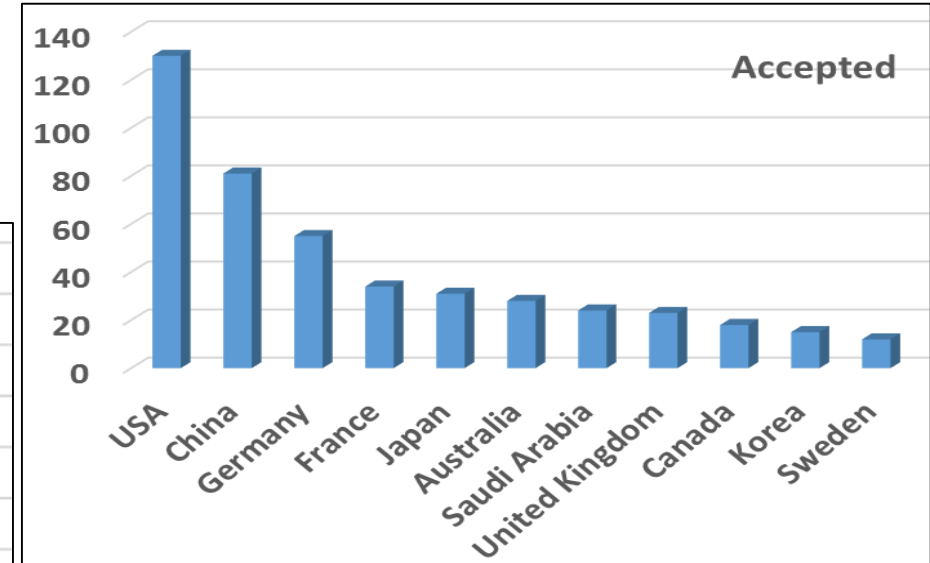
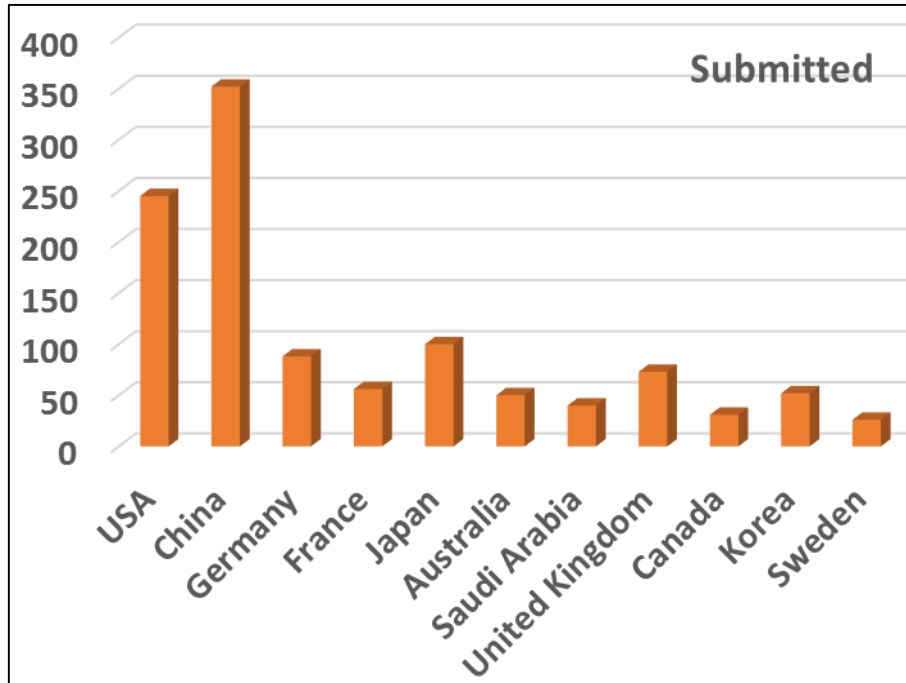
Country	Submitted	Accepted
Australia	50	28
Austria	1	0
Belgium	11	1
Brazil	6	2
Canada	31	18
Chile	6	1
China	352	81
Denmark	2	0
Egypt	3	0
France	56	34
Germany	88	55
Greece	4	1
Hungary	3	2
India	35	6
Iran	3	0
Ireland	12	6
Israel	4	1
Italy	19	8
Japan	100	31
Korea	52	15
Lebanon	1	0
Malaysia	1	0
Mexico	2	0
Netherlands	6	2

Country	Submitted	Accepted
Norway	5	3
Poland	7	1
Portugal	3	2
Qatar	1	0
Russian Federation	21	7
Saudi Arabia	40	24
Singapore	3	0
Slovenia	1	0
Spain	14	4
Sweden	26	12
Switzerland	6	6
Taiwan	16	3
Turkey	5	0
United Arab Emirates	1	0
United Kingdom	73	23
United States	245	130

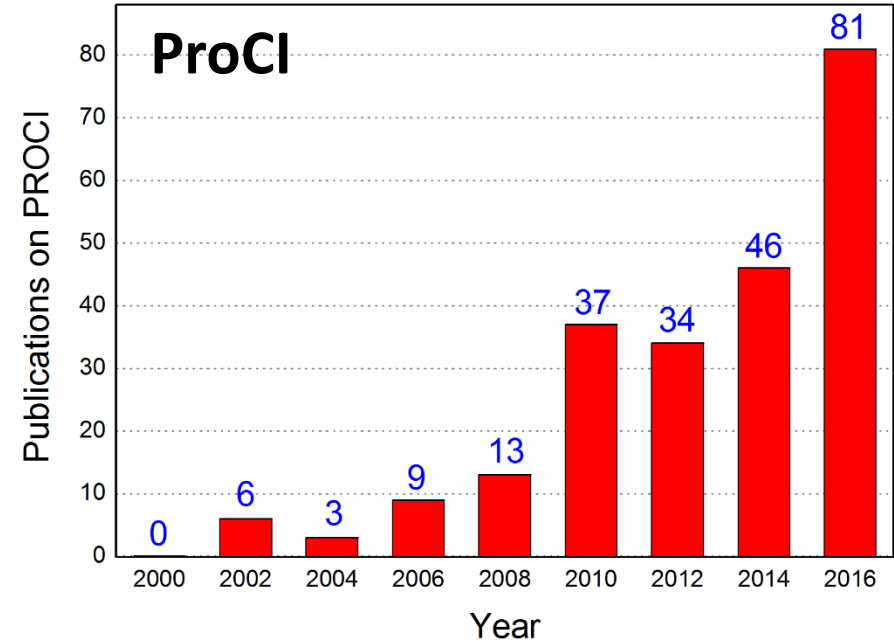
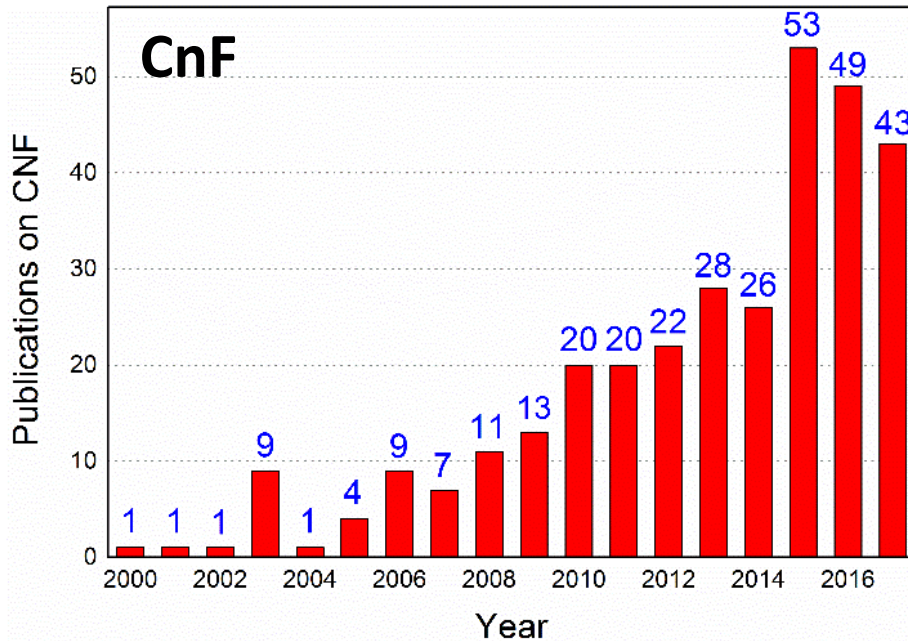
Total 507

- Black = Constant
- Red = Increase
- Blue = decrease

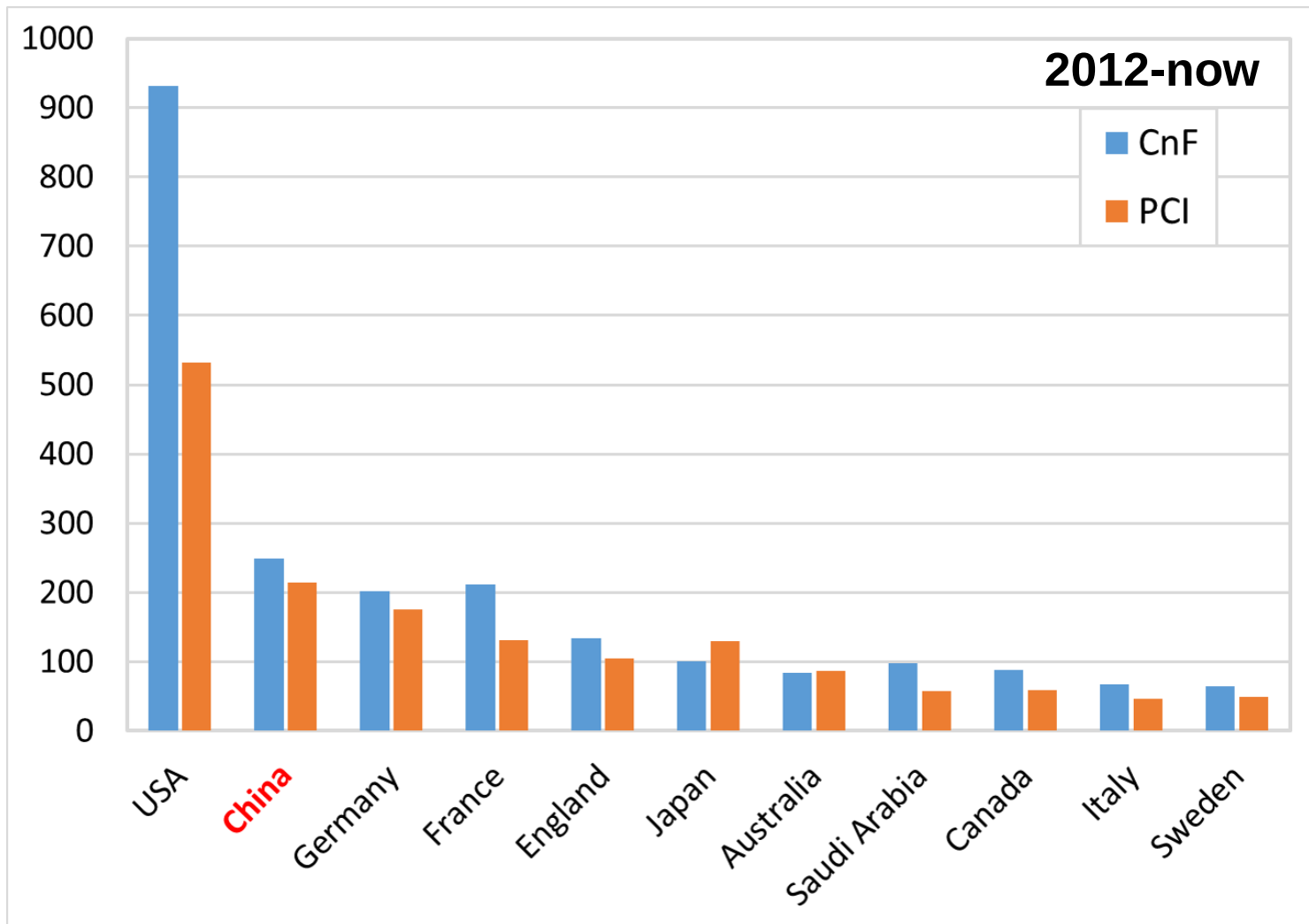
Publications — the 36th ProCI



Publications from China — CnF and ProCI



Publications from China — CnF and ProCI



Publications from China — CnF and ProCI



Publication Ratios from 2007-2017

Countries/Territories	Record Count	% of 2614	Bar Chart
CnF			
USA	1203	46.021 %	
FRANCE	290	11.094 %	
PEOPLES R CHINA	262	10.023 %	
GERMANY	247	9.449 %	
ENGLAND	202	7.728 %	
JAPAN	124	4.744 %	
CANADA	107	4.093 %	
AUSTRALIA	106	4.055 %	
SWEDEN	92	3.520 %	
ITALY	88	3.366 %	
SPAIN	80	3.060 %	
SAUDI ARABIA	79	3.022 %	
SOUTH KOREA	76	2.907 %	
RUSSIA	69	2.640 %	
NETHERLANDS	65	2.487 %	
INDIA	56	2.142 %	
IRELAND	53	2.028 %	
BELGIUM	39	1.492 %	
SWITZERLAND	39	1.492 %	

Countries/Territories	Record Count	% of 1986	Bar Chart
ProCI			
USA	903	45.468 %	
JAPAN	227	11.430 %	
GERMANY	222	11.178 %	
PEOPLES R CHINA	189	9.517 %	
FRANCE	181	9.114 %	
ENGLAND	144	7.251 %	
AUSTRALIA	91	4.582 %	
CANADA	85	4.280 %	
ITALY	77	3.877 %	
SWEDEN	63	3.172 %	
SOUTH KOREA	44	2.216 %	
NETHERLANDS	37	1.863 %	
RUSSIA	35	1.762 %	
SWITZERLAND	27	1.360 %	
SAUDI ARABIA	25	1.259 %	
SPAIN	23	1.158 %	
INDIA	22	1.108 %	
DENMARK	17	0.856 %	
IRELAND	17	0.856 %	

Homework



Some devices (i.e., Kodak film, Walkman, video player...) used to be very popular, but they are disappearing recently as the technology is developing. Which **combustion-powered device** may disappear in a decade?



Thanks for your attention

